

BASIC PROFESSIONAL TRAINING COURSE

Module **VII**

Probabilistic safety assessment



IAEA

International Atomic Energy Agency

International Atomic Energy Agency, May 2015

v1.0

Background

In 1991, the General Conference (GC) in its resolution RES/552 requested the Director General to prepare 'a comprehensive proposal for education and training in both radiation protection and in nuclear safety' for consideration by the following GC in 1992. In 1992, the proposal was made by the Secretariat and after considering this proposal the General Conference requested the Director General to prepare a report on a possible programme of activities on education and training in radiological protection and nuclear safety in its resolution RES1584.

In response to this request and as a first step, the Secretariat prepared a Standard Syllabus for the Post-graduate Educational Course in Radiation Protection. Subsequently, planning of specialised training courses and workshops in different areas of Standard Syllabus were also made. A similar approach was taken to develop basic professional training in nuclear safety. In January 1997, Programme Performance Assessment System (PPAS) recommended the preparation of a standard syllabus for nuclear safety based on Agency Safety Standard Series Documents and any other internationally accepted practices. A draft Standard Syllabus for Basic Professional Training Course in Nuclear Safety (BPTC) was prepared by a group of consultants in November 1997 and the syllabus was finalised in July 1998 in the second consultants meeting.

The Basic Professional Training Course on Nuclear Safety was offered for the first time at the end of 1999, in English, in Saclay, France, in cooperation with Institut National des Sciences et Techniques Nucleaires/Commissariat a l'Energie Atomique (INSTN/CEA). In 2000, the course was offered in Spanish, in Brazil to Latin American countries and, in English, as a national training course in Romania, with six and four weeks duration, respectively. In 2001, the course was offered at Argonne National Laboratory in the USA for participants from Asian countries. In 2001 and 2002, the course was offered in Saclay, France for participants from Europe. Since then the BPTC has been used all over the world and part of it has been translated into various languages. In particular, it is held on a regular basis in Korea for the Asian region and in Argentina for the Latin American region.

In 2015 the Basic Professional Training Course was updated to the current IAEA nuclear safety standards. The update includes a BPTC text book, BPTC e-book and 2 "train the trainers" packages, one package for a three month course and one package is for a one month course. The "train the trainers" packages include transparencies, questions and case studies to complement the BPTC.

This material was prepared by the IAEA and co-funded by the European Union.

Editorial Note

The update and the review of the BPTC was completed with the collaboration of the ICJT Nuclear Training Centre, Jožef Stefan Institute, Slovenia and IAEA technical experts.

CONTENTS

1	INTRODUCTION.....	5
2	PSA SCOPE AND LEVELS.....	7
3	FULL POWER PSA LEVEL 1 FOR INTERNAL INITIATORS.....	9
3.1	PSA Level-1 objectives.....	9
3.2	Steps of Level 1 PSA.....	9
3.3	Identification and grouping of initiating events.....	10
3.4	Accident sequence analyses.....	11
3.5	Systems analysis.....	13
3.6	Parameter estimation.....	15
3.7	Human reliability analysis (HRA).....	16
3.8	Human errors.....	17
	Pre-accident human errors.....	17
	Human errors that could lead to an initiating event.....	17
	Post-accident human errors.....	17
	Inclusion in PSA model.....	18
3.9	Model integration and quantification.....	19
	PSA model integration.....	19
	PSA model quantification.....	20
3.10	Analysis and interpretation of the PSA results.....	20
	Importance analyses.....	21
	Uncertainty analysis.....	21
	Sensitivity studies.....	23
4	PSA LEVEL 2.....	25
4.1	Objectives and general steps.....	25
4.2	Level 1 – Level 2 interface.....	26
4.3	Accident progression (APET) or containment (CET) event trees.....	27
	Quantification process for CET (APET).....	28
4.4	Physical phenomena.....	29
4.5	Management actions.....	30
4.6	Source term evaluation.....	30
4.7	Treatment of uncertainties.....	32
4.8	Presentation of Level-2 PSA results.....	32
	Containment performance results.....	32
	Source terms results.....	33
5	PSA LEVEL 3.....	34
5.1	Objectives, scope and steps of PSA Level 3.....	34
5.2	Source term.....	34
5.3	Atmospheric dispersion and deposition.....	35
5.4	Meteorological data and sampling of these data.....	35
5.5	Exposure pathways and dose assessment.....	36
	Data on population, agriculture and economy.....	37
	Counter measures.....	37
5.6	Presentation of the results.....	37
6	PSA FOR EXTERNAL AND INTERNAL HAZARDS.....	38

6.1	Definitions	38
6.2	Analysis process and collection of initial information...	38
6.3	Identification of hazards	39
6.4	Screening of hazards	40
6.5	Bounding and detailed analyses	42
7	LOW POWER AND SHUTDOWN PSA	43
7.1	General information.....	43
7.2	Plant operational states (POS).....	44
7.3	Initiating events analysis	44
	New or higher possibilities of initiators.....	45
	High potential for human-induced initiating events	45
	Initiating events grouping in POSs	45
7.4	Reliability of systems, human reliability and data analysis	45
8	PSA APPLICATIONS	48
8.1	Safety assessment.....	48
	Assessment of overall plant safety	48
	Periodic safety review	48
	Analysis of the degree of defence against assumed terrorist attack scenarios (security application).....	48
8.2	Design evaluation.....	49
	Application of PSA to support decisions made during the NPP design.....	49
	Assessment of the safety importance of deviations between an existing plant design and updated/revised deterministic design rules.....	49
8.3	NPP operation.....	49
	NPP maintenance	49
	Accident mitigation and emergency planning	50
	Personnel training	51
8.4	Permanent changes to the operating plant.....	52
	Plant changes	52
	Technical specification changes.....	52
	Establishment of graded QA programme for SSC	53
8.5	Supervisory activities	53
	Performance monitoring	53
	Performance assessment.....	54
8.6	Evaluation of safety issues.....	54
	Risk evaluation.....	54
	Regulatory decisions.....	54
9	QUESTIONS	56
10	REFERENCES	57

1 INTRODUCTION

Learning objectives

After completing this chapter, the trainee will be able to:

- 1. Describe the main purpose of performing safety analyses.*
- 2. Identify who is responsible for performing safety analyses.*
- 3. Describe the main goals/outcomes of probabilistic safety assessments.*

As already stated in Module VI, safety analyses are analytical studies aimed at demonstrating that the safety requirements are met for all possible operating conditions of a nuclear facility, in our case of a nuclear power plant, for various postulated initiating events.

This Module covers probabilistic safety assessments. The terms “safety analysis” and “safety assessment” are often used interchangeably and some languages even do not have a separate translation for these two terms. The IAEA Safety Glossary [1] acknowledges the interchangeable use of both terms but specifies that “when the distinction is important, safety analysis should be used for the study of safety, and safety assessment for the evaluation of safety”.

In this Module we will not make a distinction between these two terms and will refer to these types of analyses as Probabilistic Safety Assessments or Probabilistic Safety Analyses (PSA) interchangeably.

The Safety Fundamentals publication, Fundamental Safety Principles [2], establishes principles for ensuring the protection of workers, the public and the environment, now and in the future, from the harmful effects of ionizing radiation. As already stated in Module VI, safety analyses are undertaken as a means of evaluating compliance with safety principles and safety requirements for all nuclear facilities. They are to be carried out and documented by the organization responsible for operating the facility, are to be independently verified and are to be submitted to the regulatory body as part of the licensing or authorization process. They are systematic processes that are carried out through the lifetime of the facility to ensure that all the relevant safety requirements are met by the proposed or actual design.

The safety analysis assesses the performance of the plant against a broad range of accident conditions, many of which may never be experienced in reality, in order to obtain a complete understanding of how the plant is expected to perform in these situations.

Both types of analysis, i.e. deterministic and probabilistic safety analysis, DSA and PSA, start from a number of initiating events and analyse the response of the facility to these initiators.

Probabilistic safety analysis is a comprehensive structured approach to identifying failure scenarios and is a conceptual and mathematical tool for deriving numerical estimates of risk. Three levels of PSA are generally recognized. Level 1 comprises the analysis of plant failures leading to the determination of core damage frequency. Level 2 includes the analysis of the containment response leading, together with Level 1 results, to the determination of containment release frequencies. Level 3 includes the analysis of off-site consequences leading, together with the results of Level 2 analysis, to estimates of risk to the public and the environment.

The Specific Safety Requirement SSR-2/1 [3] on NPP Design specifies that the design must take due account of the probabilistic safety analysis of the plant for all modes of operation and for all plant states, including shutdown, with particular reference to [3]:

- Assurance that a balanced design has been achieved such that no particular feature or postulated initiating event makes a disproportionately large or significantly uncertain contribution to the overall risk;
- Assurance to the extent practicable, that the levels of defence in depth are independent;
- Assurance that small deviations in plant parameters that could give rise to large variations in plant conditions (cliff-edge effects) are prevented;
- Comparison of the results of the analysis with the acceptance criteria for risk where these have been specified.

There is an international requirement [3] adopted in almost all national legislations that both deterministic and probabilistic safety analysis must be used in the safety analysis of the design of nuclear power plants to enable the challenges to safety in the various categories of plant states to be evaluated and assessed. Such analyses are an integral part of any licensing process and part of the Final Safety Analysis Report (FSAR) for every nuclear power plant.

Both types of safety analysis support safe operation of the plant by serving as an important tool in developing and confirming plant protection and control system set points and control parameters. They are also used to establish and validate the plant's operating specifications and limits (technical specifications), normal operating procedures, maintenance and inspection requirements, emergency operating procedures (EOPs), and severe accident management guidelines (SAMGs).

2 PSA SCOPE AND LEVELS

Learning objectives

After completing this chapter, the trainee will be able to:

1. *Define the scope of probabilistic safety assessments.*
2. *Define Level 1, 2 and 3 of probabilistic safety assessments.*
3. *Name plant states for which probabilistic safety assessments are performed.*
4. *List which initiating events and hazards have to be taken into account when performing probabilistic safety assessments.*

The general objective of a PSA is to evaluate the probability of an event with serious consequences occurring at an installation. Since the first application of this approach [4], it appeared that the interest in PSA is not only to give an overall assessment of risk, but also (and mainly) to provide insights into plant design, performance and environmental impacts, including identification of the dominant risk contributors and the comparison of options for reducing risk. PSA provides a consistent and integrated framework for safety-related decision-making.

In the case of nuclear plant safety, in international practice three levels of PSA are generally recognised as defined in Ref. [5]:

- In Level 1 PSA the design and operation of the plant are analysed to identify the sequences of events that could lead to core damage and the core damage frequency is estimated. Level 1 PSA provides insights into the strengths and weaknesses of the safety related systems and procedures in place or envisaged to prevent core damage.
- In Level 2 PSA the chronological progression of core damage sequences identified in Level 1 PSA is evaluated, including a quantitative assessment of phenomena arising from severe damage to reactor fuel, and identifies ways in which associated releases of radioactive material from fuel could result in releases to the environment. It also estimates the frequency, magnitude and other relevant characteristics of the release of radioactive material to the environment. This analysis provides additional insights into the relative importance of accident prevention and mitigation measures and the physical barriers to release of radioactive material to the environment (for example, a containment building).
- In Level 3 PSA public health and other societal consequences such as the contamination of land or food from these accident sequences that lead to a release of radioactivity to the environment are estimated.

Level 1 PSA, Level 2 PSA and Level 3 PSA are performed sequentially, where the results of each assessment serve as a basis for the PSA at the subsequent level.

Ref. [5] provides recommendations towards the scope of PSA: "... in order to use the PSA results for the verification of compliance with existing safety goals or criteria, a full scope PSA involving a comprehensive list of initiating events and hazards and all plant operational modes should be performed...".

Figure 2.1 illustrates the overall possible scope of PSA. It is three-dimensional. First of all, depending on consideration of the extent of accident scenario development, Level-1, Level-2, and Level-3 PSA are distinguished. Next, the scope of initiating events and hazards may include internal initiating events caused by random components failures and/or human errors, internal hazards such as fires and floods, and external hazards both natural (such as an earthquake) and human-induced (e.g. an airplane crash). In addition, the analysis can be conducted for full power operating conditions as well as for low power and shutdown modes of the plant.

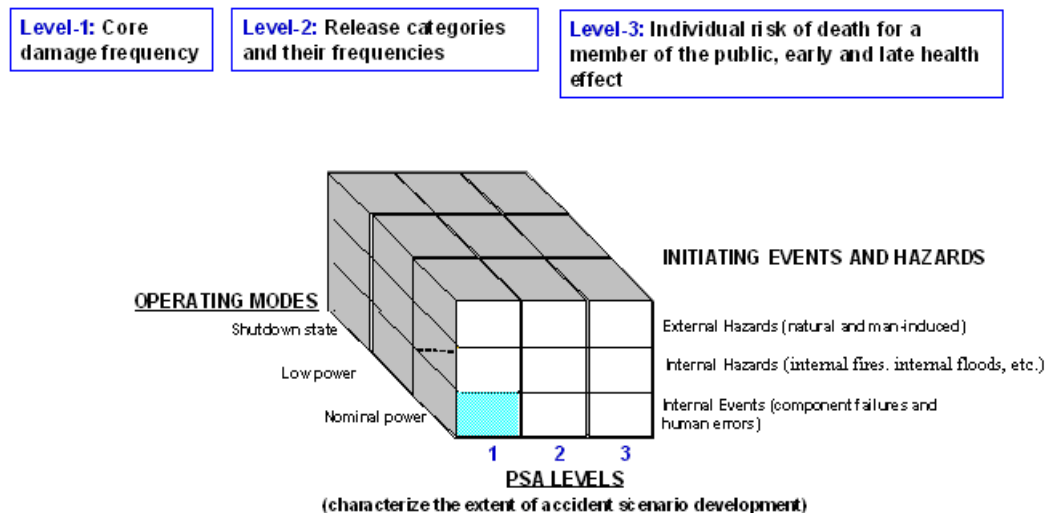


Figure 2.1: Scope of PSA.

3 FULL POWER PSA LEVEL 1 FOR INTERNAL INITIATORS

Learning objectives

After completing this chapter, the trainee will be able to:

1. Define PSA Level 1 objectives.
2. Describe the concept of fault trees.
3. Describe the concept of event trees.
4. Describe the concept of common cause failures and human reliability analysis.

3.1 PSA Level-1 objectives

Level 1 PSAs have now been carried out for most nuclear power plants worldwide. In the IAEA Safety Guide [5] it is stated: “Level 1 PSA provides insights into design weaknesses and into ways of preventing accidents leading to core damage, which might be the precursor of accidents leading to major releases of radioactive material with potential consequences for human health and the environment”.

A PSA Level 1 gives information that is very useful in analysing the safety of nuclear plants, including:

- The dominant accident sequences leading to core damage;
- Systems, initiating events, components and human actions that are important for safety and their relative importance;
- Important dependencies between systems, components, and human actions;
- Impact of different uncertainties on the estimates and insights.

PSA at nominal power for the internal initiators highlighted on Figure 2.1 usually receives the most attention, and core damage frequencies are generally less difficult to estimate than the frequencies for initiators induced by internal and external hazards. Internal initiators are typically based on operating experience and engineering analyses. In addition, the model for Level-1 PSA for internal initiators serves as a basis for PSA for hazards of other types, operation modes and levels.

3.2 Steps of Level 1 PSA

Since the aim of a PSA is to assess the probability of rare events where no direct statistics is available, the general approach is to consider the result as a combination of events with an observable frequency. More precisely, a core damage event is considered to be the result of diverse combinations of hardware and/or human failures.

The general PSA approach is then to identify all credible event sequences that could lead to core damage and to evaluate their frequency, relying on the basic event probabilities.

The steps of the study are the following (see Figure 3.1 [5, 6]):

- Identification and grouping of initiating events;
- Definition and modelling of event sequences;
- Systems analysis;
- Parameter estimation; analysis of dependent failures, common cause failures and human reliability;
- Model Integration and Quantification;
- Analyses and Interpretation of the results.

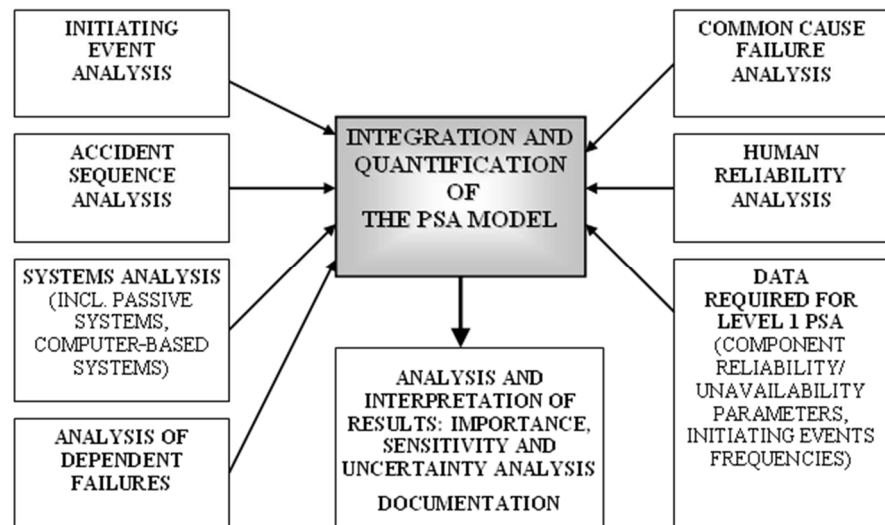


Figure 3.1: Level-1 PSA Process.

3.3 Identification and grouping of initiating events

The starting point of Level 1 PSA is identification of the set of initiating events and their grouping.

An Initiating Event (IE) can be defined as an event which creates a disturbance in the plant and has the potential of leading to core damage, depending on the successful operation of mitigating functions in the plant. For PSA purposes, the initiating events are generally classified into Internal IEs and IEs caused by Internal or External hazards [5].

Internal IEs are all plant events of internal origin due to random hardware failures or to human errors. The loss of the off-site power supply caused by reasons other than external hazards is also generally considered within the internal IEs.

The main objectives of the initiating events analysis are as follows:

- To identify a reasonably complete set of events that interrupt normal plant operation and that require successful mitigation to prevent core damage, so that no significant contributor to core damage is omitted;
- To group initiating events to facilitate the efficient modelling of plant response and assessment of the frequency of initiating events, while providing sufficient resolution regarding modelling of accident sequences (events included in the same group have similar mitigation requirements, or are bounded by the limiting mitigation requirements for the ‘representative initiating event’ for the group);
- To provide estimates of the frequencies of the initiating event groups using the information available and associated estimation techniques.

The following methods for identification of potential IEs are usually used:

- Analysis of lists of IEs from PSAs for similar units;
- Analysis of generic lists of IEs;
- Analysis of operational experience;
- Deductive analysis (e.g., master logic diagram, heat balance fault trees, etc.);
- Inductive analysis (e.g., failure mode effect analysis (FMEA));
- Lists of design basis accidents (DBAs) and design extension conditions (DEC).

IEs are grouped in separate groups with similar mitigation requirements for all IEs in the group in order to facilitate an efficient but realistic estimation of CDF (e.g., a manageable number of accident sequence models and sufficient information for their frequency estimation).

Initiating events are grouped in a single group only when the following can be assured:

- The events have the same safe and unsafe end states and lead to similar accident progression in terms of plant response, success criteria, timing, and their effect on the operability of relevant mitigating systems and operator performance; or
- The events can be subsumed into a group and bounded by the worst case impacts within the ‘new’ group.

A representative IE for further modelling of each IE group should be selected.

The different IE groups are characterized by different impacts on plant performance, safety functions, and possibilities for recovery.

3.4 Accident sequence analyses

Once initiating events have been identified and grouped, it is necessary to determine the response of the plant to each group of initiating events. This modelling of responses results in the generation of accident (or event) sequences.

The classical method for the definition of event sequences is the conventional event tree method. For reference, this is a logical method consisting of taking an initiating event and systematically envisaging the success or failure of all the systems and functions liable to play a role during the accident situation. Each branching through the tree represents an accident sequence which is then analysed to determine which branches lead to core damage, and the exact nature of the missions of the systems involved. The definition of the mission of a system constitutes the criterion of success (for example the number of trains necessary, the time of the mission, the time limit for taking human action etc.), that is to say all the information necessary for subsequently calculating the probability of failure. An example of a simple event tree consisting of an initiating event (I), modelling functions (A, B and C), and successful (OK) and unsuccessful (CD) end states is given on Figure 3.2.

The task of modelling accident sequences needs specialised efforts in very different fields, and illustrates the synthetic character of the PSA approach. Figure 3.3 shows the interrelations between an initiating event, event trees, fault trees, basic events, etc. that jointly constitute the PSA model.

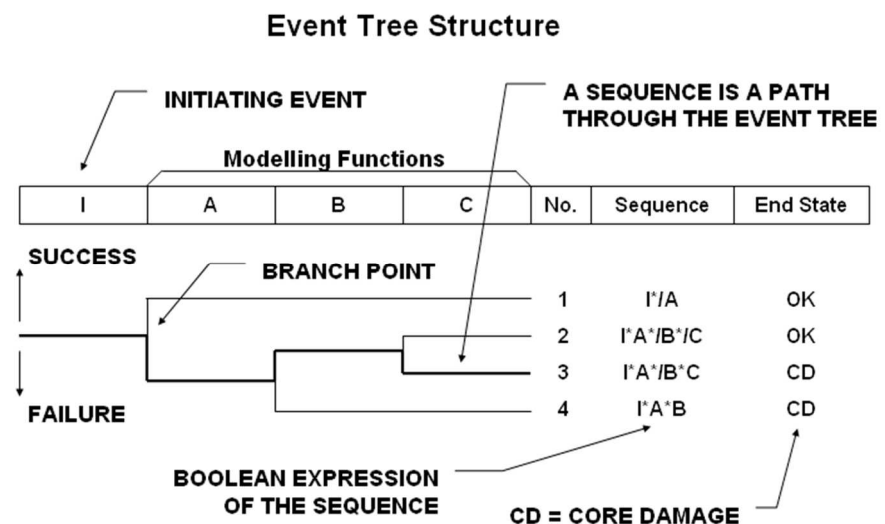


Figure 3.2: Event Tree structure.

Special efforts should be made to create scenarios that are as complete and realistic as possible. Event tree construction is a particularly important task of PSA, since it is the basis of the whole model.

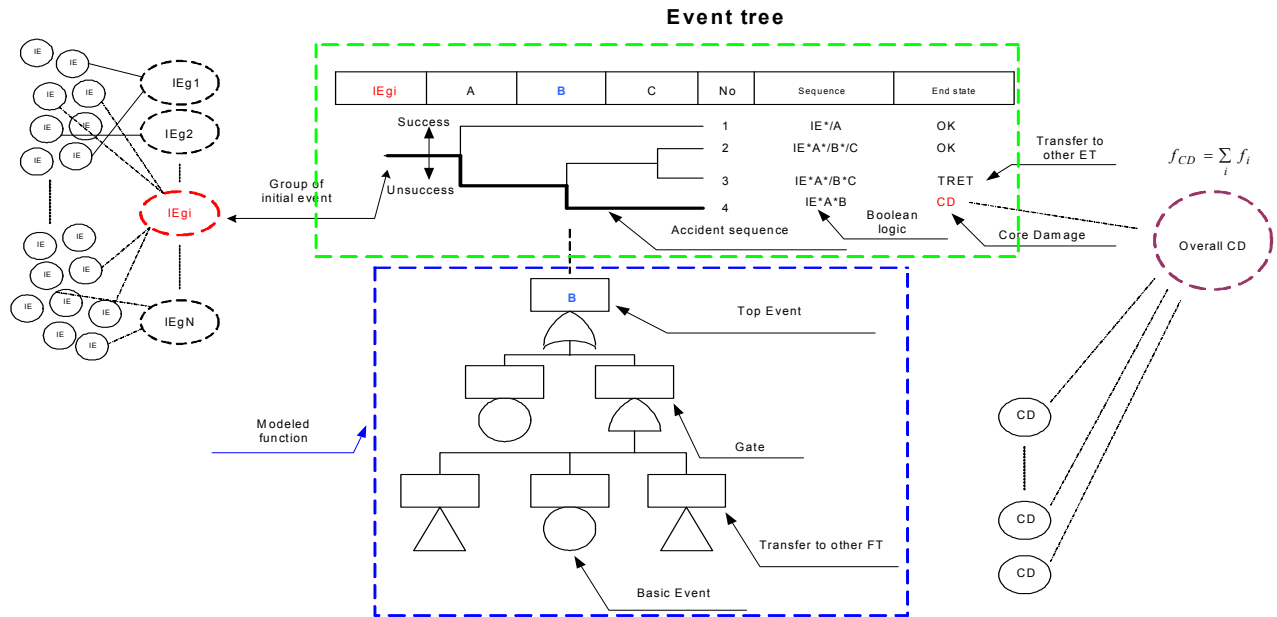


Figure 3.3: Interrelation between different PSA elements and the Event Tree.

3.5 Systems analysis

Once the response of the plant to an initiating event has been modelled, the events that constitute the event sequences need further analysis. The main events are the success or failure of the applicable systems.

The failure of the system is determined from a logical model called a fault tree (see Figure 3.4), which expresses the system's failure as a logical combination of the failures of its components and supporting systems. For a particular event called the top event (usually the failure of a system to perform some extended function), a fault tree is used to identify the combination of basic events (usually component failures or operator errors) that could lead to the top event. Due to the fact that the safety systems are usually redundant, it is important to take into account common mode failures that are simultaneous failures of multiple components due to some underlying common cause such as design errors, environmental or other factors.

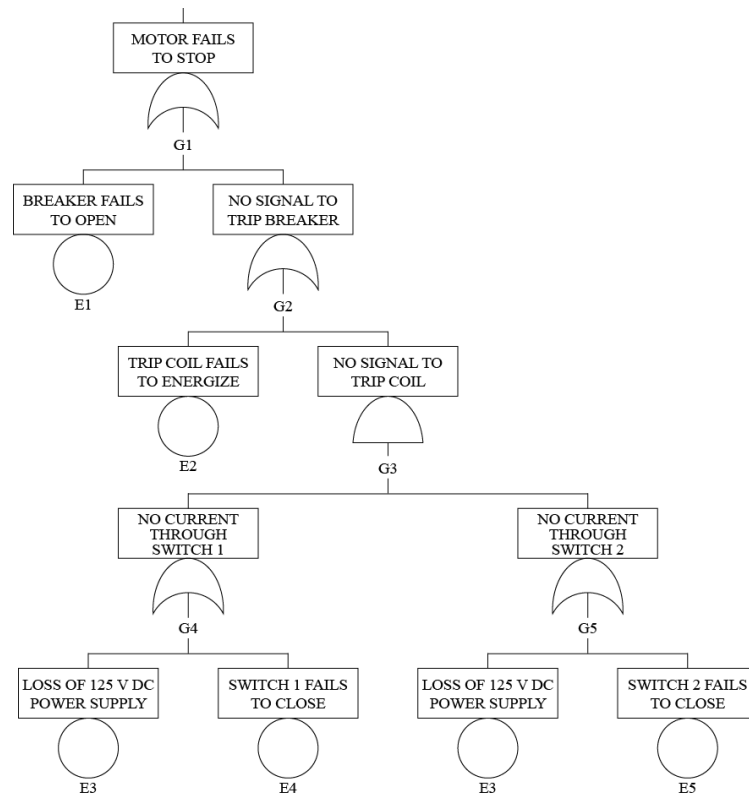


Figure 3.4: Example of a Fault Tree.

The systems analysed are involved either in accident sequences or as initiating events. According to the event sequence in which it is involved, a system may have several different missions (several different success criteria). Systems analysis and modelling has to be carried out for each mission of the system.

The first step of systems analysis is to have a very good understanding of the operation of the system, as well as the operation of its components and the effect of their failure on system success. Such an understanding can be achieved through a qualitative analysis, e.g. a failure modes and effect analysis (FMEA) [6].

After the qualitative step, system analysis is carried out using a logic model enabling calculation of the probability of failure of each mission of this system on the basis of the failure probabilities of its basic components. Several models are available (fault trees, state graphs, Petri networks).

Fault tree analysis is the most common method used for representing the failure logic of plant systems. This is deductive failure analysis, that is, an analytical technique whereby an undesired state of a system is specified, and the system is then analysed in the context of its environment and operation to find all credible ways in which the undesired state could be brought about. Techniques for fault tree construction and quantification can be found in several references, for example [6, 7].

Some operating modes of a system are more difficult to quantify with a fault tree. For example, this is the case for repairable systems or for systems with a dynamic mode of operation (their success criteria depend on time). In some cases, the use of fault trees for quantification is an approximation that could be too conservative. Other models are available, mainly state graphs (the best known is the Markov graph), or Petri-nets. Although these models allow a better representation of system operations, they are generally too complex for regular use and are limited to particular cases, such as for validating more simple models.

3.6 Parameter estimation

After the construction of logic models (event trees - fault trees), in order to quantify these models it is necessary to introduce numerical values for the different parameters.

The data necessary for quantification can be grouped into the following categories:

- Frequency of initiating events (IEs);
- Reliability/unavailability data of components;
- Common Cause Failure (CCF) data.

Frequency of IEs: As far as possible, operational experience should play a key role in quantification of the frequencies of initiating events. There are analytical approaches (Bayesian analysis) that allow plant-specific operational data to be combined with other generic information.

Reliability data of components: The reliability parameters that are used in PSAs are typically the following: a) Failure to run rate (for a component in operation); b) Failure upon demand or failure to start (for a component in stand-by); c) Failure rate during stand-by; d) Repair rate or repair time; e) Unavailability for maintenance or for testing.

As for the initiating events frequency, so operational experience should play a key role in the quantification of component failure rates or probabilities. In order to obtain specific estimates of these parameters, it is first necessary to organize data collection as completely as possible. This data collection has to contain all the information relating to failures or malfunctioning of components.

Other information also needed is total running time (or the number of demands), test intervals and test duration, unavailability for maintenance, and repair time.

If specific experience feedback is not available, other information

sources can be used, for example international data banks or data used in other PSAs. However, the applicability of such data to the plant under consideration has to be analysed. Specific data can be aggregated with generic data using Bayesian methods.

Common cause failures (CCF): CCFs are failures due to the same cause that can affect simultaneously, or within a short period of time, a number of redundant components. The common points identified (for example common support systems) are explicitly introduced in the systems models and are not considered as CCFs. The CCFs can result, for instance, from errors in design, fabrication or installation, or from the effects of the environment (normal or accident condition). A typical example would be the failure of several pumps due to inappropriate lubrication.

CCFs have to be introduced in a PSA, since their impact on system reliability is significant, especially in the case of highly redundant systems. Several methods can be used in the PSAs for CCF modelling: Multiple Greek Letters (MGL), Binomial Failure Rate (BFR), Beta Factor, or Alpha Factors [9]. The choice of model is not the most important problem for the treatment of CCFs, since with the same basic data all models give similar results. Two aspects seem more important and can be the source of large uncertainties: a) the list of components (and of failure modes) which can be submitted to a CCF (CCF component groups); b) the source of data, which is generally sparse and difficult to interpret. An example of CCF data collection is the CCF database offered by OECD/NEA [9].

3.7 Human reliability analysis (HRA)

Another type of data that is required for PSA is human reliability data. However, this data is obtained in the course of Human Reliability Analyses rather than from statistical treatment of events from plant operational experience. The objective of human reliability analysis is to incorporate into the PSA model the impact of plant personnel actions on risk.

The human errors considered are of three types [6]:

- Human errors made before the occurrence of the initiating event that have the potential to lead to the failure or unavailability of safety related equipment or systems (usually referred to as Type A human interactions). These can occur during repair, maintenance, testing or calibration tasks. If such errors remain undetected, the component or component groups affected will be unavailable when required after an initiating event.
- Human errors that could lead to an initiating event (Type B human interactions).
- Human errors that can be made by plant operators while carrying out actions after the occurrence of an initiating event

(Type C human interactions).

Important HRA tasks are:

- The identification of specific human activities whose impact should be included in the analysis;
- The representation of the impact of success in performing or failure to perform those activities correctly in the accident sequence models (e.g. event trees) and the system reliability models (e.g. fault trees);
- The estimation of the probabilities of human errors [HE]).

Analyses of the above tasks should be performed using a systematic process and in such a way that plant-specific and, where necessary, accident scenario specific factors are taken into account.

In addition, the dependence between the human failure events should be properly characterized and taken into account to ensure that the accident sequence frequency estimations are performed correctly.

3.8 Human errors

Pre-accident human errors

These errors are identified during system analysis. A systematic review of plant procedures should be carried out to identify the repair, maintenance, testing and calibration tasks carried out by plant staff for the systems modelled in the Level 1 PSA and thereby to identify Type A human interactions. The review should determine the potential for errors to occur and the effect of these potential errors on the unavailability or failure of safety systems equipment.

A well-known and appropriate approach is the THERP method [10] that provides tables for quantification. However, validation by plant-specific experience is desirable.

Particularly important are interactions that have the potential to result in the simultaneous unavailability of multiple trains of safety systems. These sources of unavailability are included in the models at component, train or system level.

Human errors that could lead to an initiating event

This category covers all the human errors that could lead to initiating events. Category B interactions are often identified by experience feedback and included in the total initiating events frequency. As a minimum, a check should be carried out to ensure that human errors that could cause initiating events are taken into account in the evaluation of the frequencies of initiating events used in the analysis.

Post-accident human errors

A systematic review of plant procedures should be carried out to

identify the actions that need to be carried out by plant operators after the occurrence of an initiating event (Type C human interactions). The review should determine the potential for errors to occur and the effect of these potential errors on the unavailability or failure of a component or system. Type C human interactions usually provide a significant contribution to the core damage frequency and hence are often the most important human interactions identified in the Level 1 PSA.

Three subcategories of Type C human interactions can be defined [6]:

- Type 1 - Procedural safety actions.
- Type 2 - Aggravating actions/errors.
- Type 3 - Recovery/repair actions.

Identification of type C human interactions needs mainly two complementary approaches:

- Analysis of the operating documents (EOPs) to identify the actions required from the operators.
- Analysis of the accident sequences to identify the key actions (positive or negative) that can influence the progress of the accident.

Type 2 actions (also called commission errors) are particularly difficult to identify. However it is recognized that important efforts should be made to identify as completely as possible the post-accident actions, including the three types of actions.

Concerning modelling and quantification, several different methods can be used and it does not seem appropriate to recommend a particular method, since none of them have been practically validated by experience. Moreover important new developments on advanced methods are presently in progress [11].

It is important to carry out a sufficiently complete qualitative analysis for identifying the main performance shaping factors (PSFs) and to use a quantitative model able to take these shaping factors into account.

Examples of shaping factors to be considered are:

- The time window available for performing an action;
- The complexity (or difficulty) of the situation;
- The organization of the operating team;
- The quality of the emergency procedures;
- The efficiency of indicators and of operator aids;
- Stress;
- Training.

Inclusion in PSA model

To represent the impact of human errors, human failure events should be either incorporated as basic events in fault trees, or used as event

tree headings.

3.9 Model integration and quantification

The main objective of the model integration and quantification process is to develop an integrated plant-specific Level-1 PSA model that will be used to estimate the core damage frequency and to develop an understanding of the contributors to the core damage model [5, 12].

PSA model integration

There are two commonly used approaches to PSA logic model construction: the so-called “Fault tree linking approach”; and the “Linked event tree approach”.

The “Fault tree linking” approach (sometimes called “Small ETs - Large FTs method”) is the most popular and many computer codes are based on its principles (e.g. SAPHIRE, RISKSPECTRUM, NUPRA). The basic principle of this method is to represent all the systems by fault trees, and to evaluate the frequency of the event sequences by linking the fault trees involved in each sequence by a logical model.

This approach has many advantages, in particular:

- An automatic treatment of the functional dependences;
- A homogeneous treatment of all the systems and sequences;
- A clear and comprehensive representation of the models;
- A reasonable calculation time, allowing for recalculations and sensitivity studies.

The major inconvenience is that the dynamic aspects cannot be treated without approximations and these may be difficult to justify.

The second approach (sometimes called “Large ETs - Small FTs method”) is also widely used (more than 30% of PSAs in the US follow this method), but only a few computer codes are based on its principles (e.g. RISKMAN). In the fault tree linking approach, at each branch point the corresponding fault tree is solved for the function or system success criteria and the boundary conditions that reflect the scenario-specific plant conditions.

Both methods produce identical results if applied properly; however, the advantages of the “Fault tree linking” approach make it more attractive for many PSA practitioners.

Other interesting methods can also be used and developments are still in progress. It is possible, for example, to use state graphs (generally Markov graphs), Petri-nets, and Monte-Carlo simulation. These methods can be used for systems modelling, as well as for sequences modelling. The main goal is to account for various specific operating

aspects. On the contrary, these methods are generally more complex, time consuming and difficult to interpret. For these reasons they are presently limited to particular cases, and especially to validating the approximations introduced in a simplified model.

PSA model quantification

The accident sequence frequencies are calculated using the data for the initiating event frequencies, component failure probabilities, component outage frequencies and durations, common cause failure probabilities and human error probabilities. The solution of the model provides logical combinations of events leading to core damage (e.g., minimal cutsets). The general structure of the integrated model is illustrated on Figure 3.3.

In accordance with [5] the overall results of the quantification of the Level 1 PSA model should include:

- The core damage frequency (point estimates and uncertainty bounds or probability distributions);
- The contributions to the core damage frequency arising from each of the initiating event groups;
- Cutsets and cutset frequencies (for the fault tree linking approach) or scenarios and scenario frequencies (for the approach using event trees with boundary conditions);
- The results of sensitivity studies and uncertainty analysis;
- Importance measures (such as the risk achievement worth and the risk reduction worth for basic events) that are used for the interpretation of the Level 1 PSA;
- Where the Level 1 PSA will be used as an input to the Level 2 PSA, the frequencies of the plant damage states that provide the interface between the two parts of the analysis.

It is essential that the check that the accident sequences or cutsets identified by the solution of the Level 1 PSA model do indeed lead to core damage in accordance with the assumptions made in the course of the development of the PSA. This check should be carried out for a sample of the sequences, focusing on those that make a significant contribution to the risk. In addition, a check should be made to confirm that the cutsets representing combinations of initiating events and component failures that are expected to lead to core damage are indeed included in the list of cutsets generated.

The Level 1 PSA documentation should present the results of the quantification of the Level 1 PSA, and should describe the most significant sequences and significant cutsets (for the fault tree linking approach) and any post-processing that has been carried out.

3.10 Analysis and interpretation of the PSA results

The objective of the results analysis and interpretation activity is to

derive an understanding of those aspects of plant design and operation that have a significant impact on risk. In addition, an important part of this task is to identify the key sources of uncertainty in the model and assess their impact on the results.

Importance analyses

Importance measures for basic events, groups of basic events, safety systems, groups of initiating events, etc. should be calculated and used to interpret the results of the PSA. The various importance measures provide a perspective on which basic events, etc. contribute most to the current estimate of risk (Fussell-Vesely importance, risk reduction worth), which contribute most to maintaining the level of safety (risk achievement worth), and for which basic events the results are the most sensitive (Birnbaum importance).

Importance measures used in Level 1 PSA typically include:

- The Fussell-Vesely importance: For a specific basic event, the Fussell-Vesely importance measure is the fractional contribution to the total frequency of core damage for all accident sequences containing the basic event to be evaluated.
- The risk reduction worth: The risk reduction worth is the relative decrease in the frequency of core damage if the probability of the particular failure mode is considered to be zero.
- The risk achievement worth: The risk achievement worth is the relative increase in the frequency of core damage if failure of a particular item of equipment is considered to be certain.
- The Birnbaum importance: The Birnbaum importance is a measure of the increase in risk when a component fails compared to when the component is operable.

Uncertainty analysis

The qualitative discussion and quantitative measure of uncertainties in the PSA results is a very important task. Uncertainty estimation is particularly important for the analysis of risk contributions, for decision-making, and for the credibility of the results.

Since the PSA model attempts to simulate reality, it is inevitable that there will be simplifying assumptions of rather complex phenomena that generate uncertainties. These uncertainties should be addressed when using the results of a PSA to derive risk insights or in support of a decision. This can be done by carrying out sensitivity studies or an uncertainty analysis as appropriate.

Three major categories of sources of uncertainties can be identified in the PSA [5]:

1. Incompleteness uncertainty: the overall aim of a Level 1 PSA is to carry out a systematic analysis to identify all the accident sequences that contribute to the core damage frequency. However, there is no guarantee that this process can ever be

complete and that all possible scenarios have been identified and properly assessed. This potential lack of completeness introduces an uncertainty in the results and conclusions of the analysis that is difficult to assess or quantify. It is not possible to address this type of uncertainty explicitly.

2. Modelling uncertainty: this arises due to a lack of complete knowledge concerning the appropriateness of the methods, models, assumptions and approximations used in the analysis. It is possible to address the significance of some of them using sensitivity studies.
3. Parameter uncertainty: this arises due to the uncertainties in the parameters used in the quantification of the Level 1 PSA. This is the type of uncertainty that is usually addressed by an uncertainty analysis through specifying uncertainty distributions for all the parameters and propagating them through the analysis.

Items (1) and (2) above are usually referred to as epistemic uncertainties (i.e. uncertainties due to lack of knowledge). Item (3) is usually termed aleatory (randomness) uncertainty.

Uncertainty analysis examines a range of alternative models or parameter values, assigns to each model or value a probability and generates a distribution of the results, within which the baseline results represent one possible outcome.

The propagation of uncertainties through the analysis can be accomplished using various methods, depending on the objective of the uncertainty analysis. Examples of available propagation techniques include: (i) the use of discrete probability distributions, and (ii) direct simulation methods based on either simple (Monte Carlo) random sampling or stratified (Latin hypercube) sampling procedures, which are primarily used nowadays.

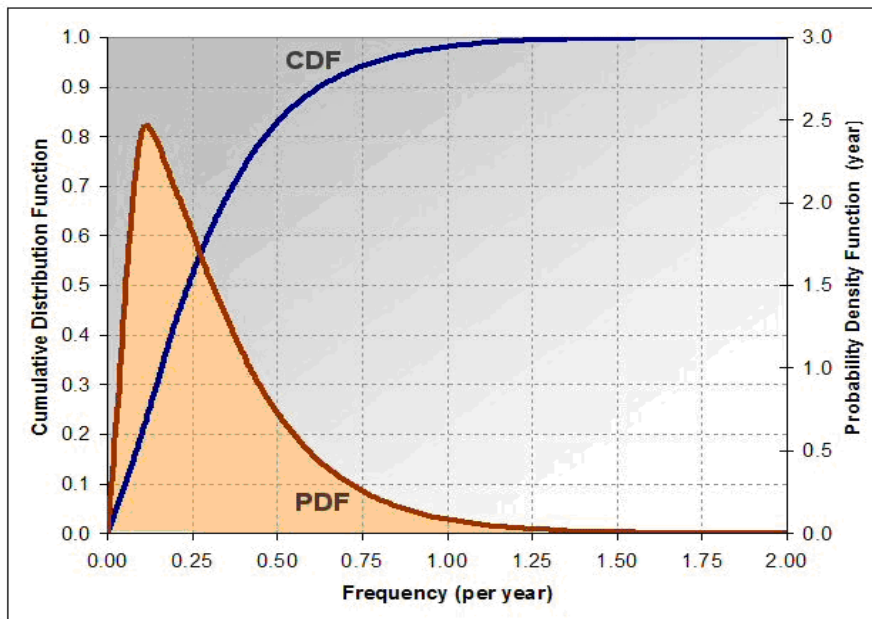


Figure 3.5: Results of the uncertainty analysis.

The results of the uncertainty analysis may be displayed using histograms, probability density functions, cumulative distribution functions (see Figure 3.5), whisker plots (see Figure 3.6), or tabular formats showing the various values of the calculated uncertainties, together with the distributional mean and median estimates.

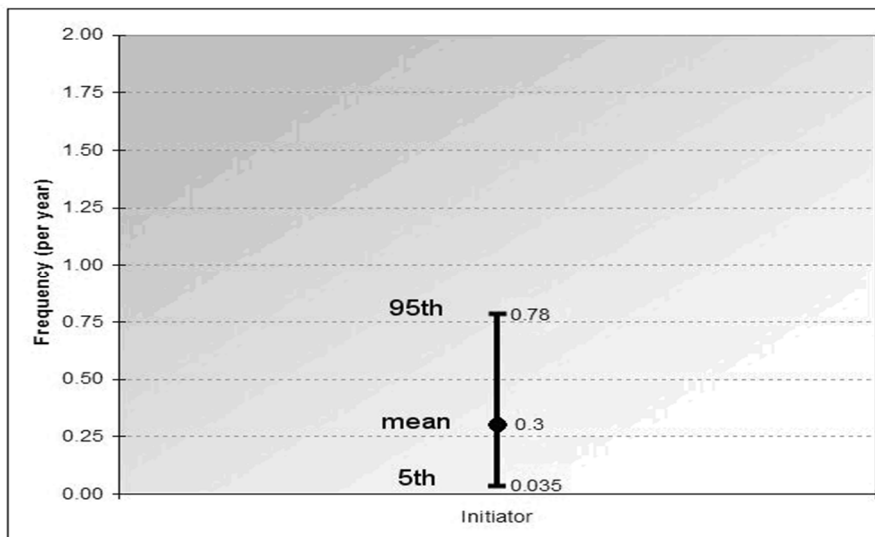


Figure 3.6: Uncertainty bounds.

Other types of uncertainty are investigated through sensitivity studies.

Sensitivity studies

Sensitivity studies should be carried out for the assumptions and data that have a significant level of uncertainty and are likely to have a significant impact on the results of the Level 1 PSA. The sensitivity

studies should be carried out by re-quantifying the analysis using alternative assumptions, or by using a range of numerical values for the data that reflect the level of uncertainty.

The results of the sensitivity studies are used to indicate the level of confidence that may be placed in the insights obtained from the PSA – that is, whether the core damage criterion or target has been met, whether the design is balanced, and whether there are possible weaknesses in the design and operation of the plant that have not been highlighted in the base case Level 1 PSA with which the sensitivity cases are compared. It should be noted that sensitivity studies are usually carried out for one assumption or one parameter at a time and that the results of the sensitivity studies have no statistical significance.

4 PSA LEVEL 2

Learning objectives

After completing this chapter, the trainee will be able to:

1. *Define PSA Level 2 objectives.*
2. *Describe the general steps undertaken when performing Level 2 PSA.*
3. *Explain the PSA Level 1 to PSA Level 2 interface.*
4. *Describe the principles of calculating accident progression and containment analyses in PSA Level 2.*
5. *Describe the key phenomena to be considered in Level-2 PSA.*

4.1 Objectives and general steps

Level 2 PSAs are devoted to a probabilistic evaluation of the containment performance and radioactive releases into the environment (source terms) for severe accidents. They are based on the results of Level 1 PSAs, dealing with core damage frequencies, and provide inputs for Level 3 PSAs, aiming at assessing the radiological and economic consequences of severe accidents.

The overall objectives of the Level 2 PSA include the following [13]:

- To gain insights into the progression of severe accidents and the performance of the containment;
- To identify plant-specific challenges and vulnerabilities of the containment to severe accidents;
- To provide an input into the resolution of specific regulatory concerns;
- To provide an input into determining compliance with the probabilistic safety goals, or with probabilistic safety criteria if these have been set. Typically such probabilistic safety goals or criteria relate to large release frequencies and large early release frequencies;
- To identify major containment failure modes and their frequencies and to estimate the associated frequencies and magnitudes of radionuclide releases;
- To provide an input into the development of strategies for off-site emergency planning;
- To evaluate the impacts of various uncertainties, including uncertainties in assumptions relating to phenomena, systems and modelling;
- To provide an input into the development of plant-specific accident management guidance and strategies;
- To provide an input into determining plant-specific options for risk reduction;
- To provide an input into the prioritization of research activities for minimization of risk significant uncertainties;
- To provide an input into Level 3 PSA consistent with the PSA

objectives;

- To provide an input into the environmental assessment for the plant.

A Level 2 PSA comprises three main steps:

- The interface with Level 1 PSA;
- The representation of possible accident progressions after core melt using an event tree, and
- The evaluation of source terms.

The quantification of the event tree and of the source terms is based on the results of supporting studies dealing with physical phenomena, component reliability and human actions.

4.2 Level 1 – Level 2 interface

The Level 1 PSA identifies a large number of accident sequences that lead to potential core damage. It is neither practical nor necessary to treat each one of these individually when assessing accident progression, containment response and fission product release. Information about the many different accidents that lead to core damage is passed from the accident sequence analysis to the accident progression by means of Plant Damage States (PDS). The sequences are grouped into Plant Damage States (PDS), so that all accidents within a given PDS can be treated as a group for the purposes of Level 2 assessments.

Broadly, PDSs can be grouped into two main classes: those in which radioactive products are released to the containment, and those in which the containment is either bypassed or ineffective.

For the first class, the PDSs should account for those plant failures defined in the Level 1 analysis that could influence either the containment challenge or the release of fission products. The grouping of sequences into PDSs can be performed according to the values of specific attributes which have a significant influence on the accident progression. This can include, for instance, the type of initiator, the timing of core damage, and/or the circuit pressure at vessel failure. The status of the containment's engineered safety features (spray system or fan coolers) and the status of other systems, such as the electrical power, are also of vital importance.

For the second class, the interface should focus on the attributes that influence the fission product source terms.

In order to reduce the number of PDSs to be considered, either one PDS can be grouped with another one that can lead to higher consequences (this will introduce additional conservatism into the process), or a frequency cut-off can be used as a means of screening

out less important PDSs (e.g., low occurrence frequency events are ignored).

4.3 Accident progression (APET) or containment (CET) event trees

In Level 2 PSAs, event trees are used to delineate the sequence of events and severe accident phenomena after the onset of core damage that challenge successive barriers to radioactive material release. They provide a structured approach for the systematic evaluation of the capability of a plant to cope with severe accidents. Their use is shown in Fig. 4.1 [6]. Such event trees are termed accident progression event trees (APET) or containment event trees (CET). The term containment event tree is adopted in most Level 2 PSAs, while accident progression event tree is less frequently used.

The top events or nodal questions in a CET (APET) address the events and physical processes that govern accident chronology, plant response to beyond design basis conditions, relevant challenges to barriers to radioactive material release and the eventual magnitude of the release of radioactive material to the environment. Nodal questions of the containment event tree also address issues and actions relating to severe accident management. The nodal questions of the CET (APET) are strongly specific to plant type, i.e. issues of importance to severe accident behaviour in one type of reactor and/or containment system may not be important to others.

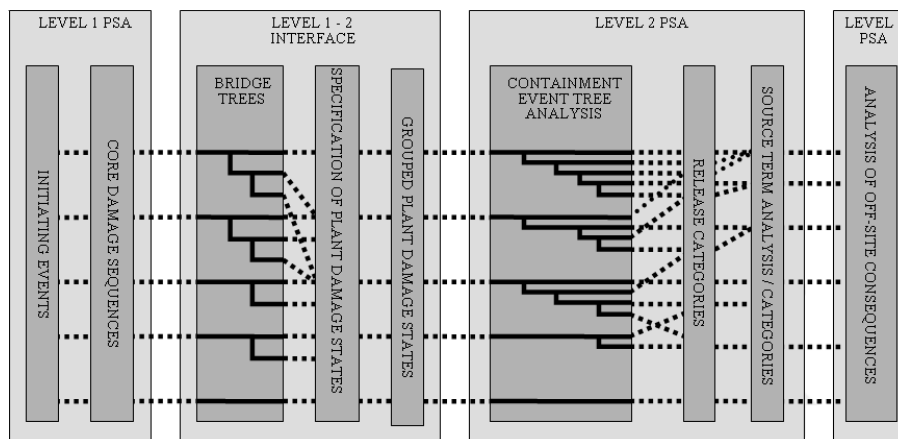


Figure 4.1: Overview of the development of a typical Level 2 PSA.

The list of events and processes addressed in the event tree can be rather extensive. Therefore, CET (APET) can grow into rather large and complicated logic models. However, relatively simple logic models can be sufficient for certain applications. Thus, for instance, if the objective of the Level 2 PSA is solely to determine the large early release frequency and a quantitative assessment of the full range of severe accident source terms is not required, smaller event tree

structures can be developed that focus on severe accident sequences with high consequences within the appropriate time frame. In any case, the overall structure of the model should be traceable by independent reviewers and manageable by the PSA team. Therefore, in the CET (APET) structures, a reasonable balance between modelling detail and practical size should be achieved.

The event tree structure should be chronologically correct, should properly take into account interdependences among events and/or phenomena, and should reflect an appropriate level of detail to satisfy the objectives of the Level 2 PSA.

Regarding chronology, it is both useful and common practice to divide the containment event tree into sequential phases in time, with the transitions between phases representing important changes in the issues that govern accident progression, such as:

Phase 1: Immediate response of the plant to the plant damage state caused by the initiating event through the early period of in-vessel core damage.

Phase 2: Late period of in-vessel core damage up to failure of the reactor pressure vessel.

Phase 3: Long term response of the plant. This phase is sometimes further subdivided into three sub-phases:

- Phase 3a - close to the time of reactor pressure vessel failure (to address challenges occurring due to failure of the reactor pressure vessel, e.g. direct containment heating);
- Phase 3b - up to a few hours after failure of the reactor pressure vessel (to address immediate ex-vessel molten core behaviour, e.g. stabilization of the melt ex-vessel or onset of the core-concrete interaction); and
- Phase 3c - long term, starting from a few hours after failure of the reactor pressure vessel (to address challenges arising from ex-vessel melt behaviour, e.g. pressurization due to the generation of non-condensable gases during the core-concrete interaction, or combustion phenomena or pressurization due to ongoing steam generation).

The CET (APET) provides the conditional probability that a containment failure mode can be realized, given a PDS. The CET (APET) produces a large number of end states, which are generally grouped in release categories or bins. One bin is defined by the values taken by several indicators that will give all the information needed to determine the release of fission product to the atmosphere.

Quantification process for CET (APET)

The assignment of conditional probabilities to branches of the containment event tree should be supported by documented analyses

and data to provide a justified representation of the uncertainty in the outcome at each node [14 - 19]. Several sources of current and relevant information can be used to support the assignment of probabilities, such as:

- a) Deterministic analyses using established computer codes for modelling severe accidents or basic principles;
- b) Relevant experimental measurements or observations;
- c) Analyses and findings from studies of similar plants;
- d) Expert elicitation involving independent experts.

For some phenomena, like direct containment heating, production of hydrogen, and steam explosions, the answers are expressed by probability distributions generally produced by expert judgment. In CET (APET) the effect of the environmental conditions resulting from a severe accident on the survivability of active components must be considered (pressure and temperature, radiation, hydrogen combustion, aerosol loading, etc.). Dependencies relating to system availability should also be correctly taken into account in the CET (APET).

4.4 Physical phenomena

All physical phenomena involved in severe accidents should be considered as far as they influence the challenge to containment integrity and the fission product source terms. A typical list of the physical phenomena to be considered is the following:

- Core heat-up, degradation and damage, cladding rupture, hydrogen production by cladding oxidation, and downflow of corium to the vessel bottom;
- Interaction between corium and water in the vessel bottom and in the reactor cavity, in-vessel and ex-vessel steam explosions, blast loads, missile generation, vessel head rupture, interaction between corium and concrete, and non-condensable gas generation;
- Containment pressurization, hydrogen combustion in the containment (laminar and turbulent deflagration, detonation), and consequences for containment integrity.
- Vessel bottom rupture, corium dispersal into the containment building, direct containment heating, and consequences for containment integrity.

A compilation of relevant severe accident phenomena can be found in Refs [20, 21]. Developments have taken place in a number of areas, such as:

- In-vessel steam explosions (alpha mode containment failure), e.g. Ref. [17];
- Direct containment heating, e.g. Ref. [20];
- Failure of the lower head of the reactor pressure vessel, e.g. Refs [21, 22];

- Flame acceleration and the transition from deflagration to detonation, e.g. Ref. [23].

Physical phenomena must be evaluated using appropriate computer codes, where the phenomena and the events that may appear in the course of the accident are modelled and where the interactions between these phenomena are correctly considered.

4.5 Management actions

It is important that actions for management of severe accidents are reflected in the Level 2 PSA model. Actions that are demanded soon after the onset of core damage can be represented in the accident sequence event trees in the Level 1 PSA model, if the conditions for their implementation can be predicted with confidence. Relevant severe accident management actions that are not represented in the Level 1 model should be incorporated into the CET (APET). Typically, such actions would be those which are expected later in the chronology of the severe accident sequence, for example, refilling of steam generators to reduce releases to the environment via damaged steam generator tubes and restarting the low pressure injection after a high temperature induced break in primary circuit boundaries. It is important to ensure that potential dependencies between operator actions included in the accident sequence models in Level 1 PSA and in the CET (APET) for Level 2 PSA are assessed and taken into account, as appropriate. Also potential adverse effects of severe accident management actions need to be considered as part of the event tree logic. (e.g. injection of water into a degraded core may be able to arrest the progression of a severe accident; however, there is also the potential for energetic fuel-coolant interaction, fuel shattering and additional releases of steam, hydrogen and radioactive material).

4.6 Source term evaluation

The different steps of source term evaluation are the following:

- Grouping of fission products in accordance with their common chemical and physical characteristics;
- Assessment of the release of fission products from fuel during the in-vessel phase;
- Consideration of the retention of fission products within the reactor coolant system;
- Assessment of the release of fission products during the ex-vessel phase;
- Consideration of the retention of fission products inside the containment.

The CET (APET) will provide the conditional probability that a containment failure mode can occur, given a PDS. The CET (APET)

produces a large number of end states, some of which are either identical or similar in terms of key release attributes. The end states are often grouped in release bins, referred to as release categories. These release categories group CET (APET) end states that would be expected to have similar radiological characteristics and potential off-site consequences.

The source term information that the Level 3 PSA requires for each release category covers:

- The radionuclides, including also the chemical form of each radionuclide;
- The frequency (or frequency distribution) of each release category;
- The amount of radionuclides released as a function of time, expressed as fractions of the initial core inventory for each group of radionuclides having similar physical and chemical characteristics;
- The time of release, relative to reactor shutdown;
- The location of release, relative to ground level;
- The energy content of release, which is a function of containment temperature and pressure prior to failure;
- The particle size distribution of released aerosols, which affects deposition during plume transport.

In order to assess these characteristics, several plant specific source term calculations should be performed with an appropriate computer code (for example Source Term Code Package – STCP, consisting of MARCH3, TRAP-MELT3, VANESA and NAUA/SPAC/ICEDF codes [23]), considering the following phenomena:

- Release of fission products during core heat-up, degradation and damage, chemical reactions between fission products, and cladding and control rods;
- Retention of fission products within the reactor coolant system, condensation of fission product vapour on structures, aerosol agglomeration and deposition, chemical interactions with metallic surfaces, re-vaporization, and mechanical re-suspension of material previously deposited;
- Release of fission products during the ex-vessel phase due to steam explosion or core concrete interaction;
- Retention of fission products inside the containment, aerosol agglomeration and deposition, chemical reactions with structures and paints (especially for iodine), effect of the spray system, and deposition inside suppression pool water (for BWRs);
- Retention of fission products within the ventilation systems and the containment filtered venting system, if such a device is installed.

4.7 Treatment of uncertainties

Similar to Level-1 PSA, uncertainties will arise in the Level 2 PSA analysis as the result of the same factors.

Loss of details due to aggregation. Grouping accident sequences or cutsets from the Level 1 PSA into plant damage states for input into the Level 2 PSA for practical reasons also introduces uncertainties due to the resulting loss of some modelling detail. Further, the process of ‘binning’ (or grouping) accident sequences introduces uncertainty through the possibility that the attributes used by the analyst to group ‘similar’ accident sequences are incomplete.

Similar to Level-1 PSA the Level 2 PSA analysts should identify the dominant sources of uncertainty in the analysis and should quantitatively characterize the effects of these uncertainties on the baseline (point estimate) results. This is typically accomplished using sensitivity analysis and uncertainty analysis as in Level 1 PSA.

The sources of uncertainty in a Level 2 PSA are numerous and it is impractical to address all of them quantitatively. Experience in performing uncertainty studies for limited aspects of severe accident phenomena suggests that the effects of uncertainties from some sources are larger and more dominant than the effects of uncertainties from other sources. In an integral sense, then, the aggregate uncertainty in Level 2 PSA results can be estimated by selecting the dominant sources of uncertainty and treating them in detail. Reference [24] provides information on an evaluation of uncertainties in relation to severe accidents and Level 2 PSA.

4.8 Presentation of Level-2 PSA results

Quantitative measures of a plant’s response to severe accidents include:

- Containment performance:
 - Conditional probability of containment failure by “mode” (time, size & location).
- Source term:
 - Probability of exceedance for source terms of various magnitude,
 - Frequency of a release greater than a criterion (For example, “large early” release as defined in by regulators in the USA or the U.K).

Containment performance results

Figure 4.2 shows an example of the presentation of the results of containment performance. The breakdown of various containment failure modes could be provided:

- Across total core damage frequency;
- For each PDS;
- For a specific accident sequence.

Source terms results

Figure 4.3 presents an example of the presentation of the results of source terms analysis.

There are many ways to describe source term results:

- As the frequency of exceeding different release magnitude (see Figure 4.3);
- As the conditional probability of a “large” release (for each PDS or for a specific containment failure mode).

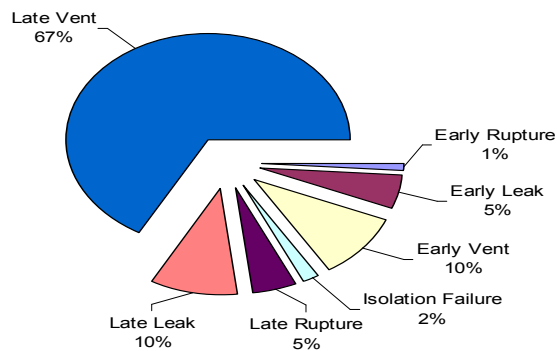


Figure 4.2: Containment Performance Results.

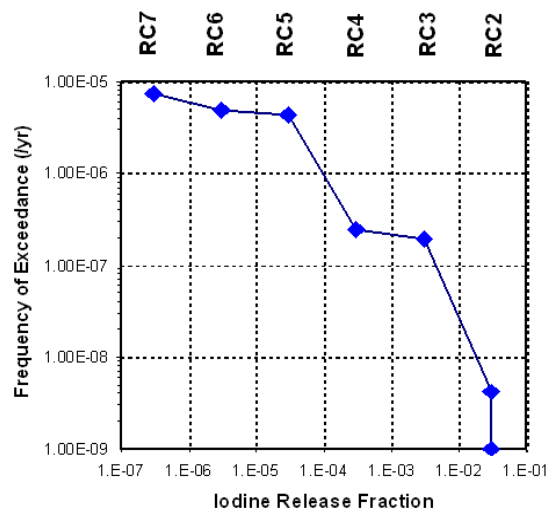


Figure 4.3: Source terms results.

5 PSA LEVEL 3

Learning objectives

After completing this chapter, the trainee will be able to:

1. Define PSA Level 3 objectives.
2. Describe the main steps needed for performing PSA Level 3.
3. Understand the concept of atmospheric dispersion and deposition.
4. Describe the use of meteorological data, exposure pathways, and data on population, agriculture and economy for dose assessment.

5.1 Objectives, scope and steps of PSA Level 3

A PSA Level 3 is needed when probabilistic public safety criteria have been defined by regulatory bodies.

The most commonly used Level 3 safety criteria are: individual risks of early and late fatal health effects; societal risk of early fatal health effects; collective risks and the risk of unacceptable land contamination.

Starting ideally from postulated releases (source terms for the environment) for a set of accidental sequences as provided (with their probabilities) by the Level 2 PSA, Level 3 PSA leads to the probabilistic accident consequence assessment, expressed in terms of risk to the public (radiological and economic consequences). Due to the current state of the art in this area, releases to the atmosphere are most frequently considered, though aquatic contamination is currently receiving more attention.

The different steps in evaluation of risks are the following [25]:

- Description of radionuclide release (including frequency) from PSA Level 2 (source term);
- Atmospheric dispersion and deposition;
- Local and global meteorological data;
- Exposure pathways;
- Population, agricultural and economic data;
- Countermeasures (if any);
- Health effects.

5.2 Source term

The expression “source term” (for the environment) covers all the characteristics of the release (for each accident sequence or for each group of sequences) that have to be known to calculate all the radiological consequences due to the different exposure pathways. These characteristics are:

- Time of the release after the beginning of the accident;
- Warning time (time available for possible implementation of protective actions before the release);
- Kinetics of the release for each radionuclide (release rate as a function of time);
- The location of the release (stack, leaks from a building with building wake effect);
- The energy associated with the release for a possible plume rise effect (depending also on meteorological conditions);
- Physical (particle size) and chemical forms (especially for iodine) for each radionuclide; this information, useful for atmospheric dispersion (deposition rates) and internal dose calculation (classes of solubility for pulmonary tract), is seldom available.

The best way to take into account the source term as a function of time, is to separate the release into a number of puffs, each of them carrying a fraction of the release for each radionuclide.

5.3 Atmospheric dispersion and deposition

Gases, vapours and aerosols are carried by the wind and during transport, the puffs (or the plume which is the global representation of the successive puffs) grow vertically and horizontally due to atmospheric turbulence, and it is generally assumed that the puffs have a Gaussian profile, characterized by standard deviations (depending on meteorological condition and on distance). This assumption has been found to be sufficiently valid in many situations and also is easy to model; transport and diffusion processes are summarized by the expression atmospheric dispersion. With an atmospheric dispersion model, for specified meteorological conditions (possibly varying with time), it is possible to calculate, at any location, the airborne concentrations and the deposited activities due to dry (surface effects due to impaction, diffusion, absorption by vegetation), and wet (due to rain falling through the puffs) deposition.

The content of each puff (for each radionuclide concerned) is depleted by radioactive decay and by deposition, depending on transfer time and meteorological conditions.

5.4 Meteorological data and sampling of these data

To obtain a probabilistic assessment of the radiological consequences, it is necessary to take into account all the various meteorological conditions during the release. So, for a specific site, it is necessary to have representative meteorological data recorded on an hourly basis for at least one year. These data are obtained at the site itself or near the site by a station of the meteorological network. For long distance

calculations and for dispersion models more sophisticated than the Gaussian one, it is necessary to have recorded meteorological data at several locations around the site, to take into account topographic effects, for example.

The meteorological data, for each hour, include:

- The wind speed and the wind direction;
- The intensity of rain (or indication of no rain);
- An indicator of atmospheric turbulence, generally in the form of a “stability class”, such as the classical Pasquill-Gifford classes from A (unstable) to F (very stable).

Different techniques of sampling these data, for the release sequences, are currently used:

- Each hourly condition can be the starting point of the release;
- Random sampling of the starting point of the release;
- Cyclic sampling (with a chosen time interval);
- Stratified sampling, where the meteorological sequences are grouped into a limited number of categories (a less time consuming technique).

5.5 Exposure pathways and dose assessment

The main exposure pathways during and after the atmospheric dispersion are:

- External exposure from the passing puffs (or the passing plume);
- External exposure from material deposited on the ground;
- Internal exposure by inhalation during the passage of the puffs (generally committed doses to various organs and effective dose);
- Internal exposure by ingestion of foodstuffs contaminated by dry and wet deposition (generally committed doses to various organs and effective dose), firstly by direct contamination and afterwards by root uptake.

For each pathway, dose coefficients (or dose conversion factors) published by national institutes or by international organizations (like ICRP or IAEA) that allow conversion, for each radionuclide, of the concentration (airborne, deposited, contained in a plant, etc.) to individual doses.

For external exposure and for people indoors, shielding factors can be taken into account. More generally, the effects of protective actions (like evacuation, sheltering, stable iodine administration, food ban, etc.) which can be implemented according to emergency plans and to the warning time, can be taken into account in the dose calculations for various time intervals after the beginning of the accident.

Data on population, agriculture and economy

These data are necessary to evaluate the health effects, and to calculate the economic impact of implementing countermeasures like relocation of people and food bans. Generally these data are available in the form of a radial grid centred on the site.

Counter measures

Short-term countermeasures include sheltering, evacuation, and administration of stable iodine. Long-term countermeasures include relocation of people, decontamination of soils (urban and rural areas) and food bans for various period of time.

5.6 Presentation of the results

Generally, the consequences are evaluated, for each release sequence and for each meteorological sequence, on all the point of a radial grid and this type of calculation is repeated for a sufficiently large number of meteorological sequences in order to take into account all the possible conditions, including the worst. All the results are ordered to obtain a relative frequency distribution and then the final results are presented as a complementary cumulative distribution function (CCDF) - not to be mistaken with the conditional core damage frequency (CCDF) used in PSA Level 1. A point on a CCDF curve gives the conditional (if the accident occurs) probability that the corresponding consequence value is not exceeded. Generally several percentiles (for instance 50 or median, 90, 95 and 99), being easily derived from the CCDF, are chosen to represent the consequences on a probabilistic basis.

6 PSA FOR EXTERNAL AND INTERNAL HAZARDS

Learning objectives

After completing this chapter, the trainee will be able to:

1. Define internal and external hazards.
2. Describe the principles of hazard identification and evaluation.

6.1 Definitions

Apart from random component failures and human errors that may lead to internal initiating events, fault sequences may be caused by the damage imposed by other hazards. Other hazards can be categorized as follows [5]:

- Internal hazards originating from sources located on the site of the nuclear power plant, both inside and outside plant buildings; examples of internal hazards are internal fires, internal floods, turbine missiles, on-site transportation accidents and releases of toxic substances from on-site storage facilities.
- External hazards originating from sources located outside the site of the nuclear power plant; examples of external hazards are seismic hazards, external fires (e.g. fires affecting the site and originating from nearby forest fires), external floods, high winds and wind-induced missiles, off-site transportation accidents, releases of toxic substances from off-site storage facilities and severe weather conditions.

Such hazards can damage plant components and thus generate accident sequences that might lead to core damage (or to other end states as appropriate, if these are to be considered in the Level 1 PSA). Often these hazards have the potential to affect many different pieces of equipment simultaneously and adversely impact plant personnel. Both internal and external hazards should be included in the Level 1 PSA (See Figure 2.1).

6.2 Analysis process and collection of initial information

The main stages of the analysis of internal and external hazards typically include:

- Collection of initial information on internal and external hazards;
- Hazard identification, including single and combined hazards;
- Hazard screening analysis, both quantitative and qualitative;
- Bounding assessment; and/or
- Detailed analysis.

While the stages of hazard identification and screening are similar for internal and external hazards, the bounding assessment and detailed analysis for each hazard may involve tasks that may be unique for the hazard considered, e.g. fire propagation will need to be analysed in the case of internal fires.

All potential internal and external hazards that may affect the plant should be considered and should be subjected to screening analysis, bounding assessment or detailed analysis, as appropriate.

As the starting point of PSA for internal and external hazards, all available information specifically relating to such internal and external hazards is collected. This information includes, as a minimum:

- Design information relating to internal and external hazards as considered in the safety analysis report;
- Lists and layouts of plant buildings, structures, systems and components;
- Plant layout and topography of the site and surroundings;
- Information on the location of pipelines, transportation routes and on-site and off-site storage facilities for hazardous materials;
- Location of industrial facilities in the vicinity of the site;
- Historical information on the occurrence of any internal and external hazards at the site, in the region, etc.

6.3 Identification of hazards

The task of hazard identification is aimed at generating a comprehensive list of potential internal and external hazards. Examples of specific hazards are:

Internal hazards inside plant buildings:

- Internal fires;
- Internal floods;
- Internal missiles;
- Internal explosions;
- Heavy load drops.

External natural hazards:

- Seismic hazards;
- External fires;
- External floods;
- High winds;
- Biological phenomena, for instance abnormal fish population in the cooling pond;
- Extreme meteorological conditions (e.g. extreme temperature, extreme atmospheric moisture, snowfall, icing, lightning etc.).

External man-made hazards:



- Off-site explosions;
- Off-site toxic substances releases;
- Aircraft crashes.

To ensure that the process of identification of hazards is comprehensive and traceable, a two-step approach should be applied as follows:

- Utilising the analysis of internal and external hazards available internationally. As a starting point, the hazards listed in various related IAEA publications (e.g. Ref. [5]), and examined in past studies should be included in the list.
- Identification of site-specific and plant-specific internal and external hazards in a structured framework that makes possible a comprehensive check.

For existing plants an integral part of the process of identification of internal and external hazards is the site survey and plant walk-down.

Combined hazards

The process of identification of hazards should include the identification of all credible combinations of hazards that may be significant for risk. Note that combinations of hazards may have a significantly higher impact on plant safety than each individual hazard considered separately, and the frequency of occurrence of a combination of hazards may be comparable to that of the individual hazards, e.g. high level water due to storm precipitation and dam failure caused by storm precipitation.

The following causes for combinations of hazards should be considered:

- Hazards having the potential to occur under the same conditions and at the same time (e.g. high winds and snowfall);
- One external hazard can induce other (e.g. a seismically induced external flood accompanied by dam failure);
- External hazards can induce internal hazards (e.g. seismically induced internal fires or floods);
- One internal hazard can induce another internal hazards (e.g. internal floods induced by internal missiles).

The impact of combinations of hazards on safety functions should be reassessed as they may affect different safety functions or the same function in a more severe manner than a single hazard.

6.4 Screening of hazards

A successive screening process is generally established to minimize the emphasis on internal and external hazards whose significance to risk is low and to focus the analysis on hazards that are risk significant. The successive screening process should be applied

consistently and the screening criteria should be specified in a manner that ensures that none of the significant risk contributors from any internal or external hazard relevant to the plant and the site are omitted.

The following screening criteria, which can be used either individually or in combination, are typically applied:

- Based on qualitative arguments, the hazard will not lead to an initiating event. For external hazards, this criterion is generally applied when the hazard cannot occur close enough to the plant to affect it. Satisfaction of this criterion will also depend on the magnitude of the hazard.
- The hazard will be slow to develop and it can be demonstrated that there will be sufficient time to eliminate the source of the threat or to provide an adequate response.
- The hazard is included within the definition of another hazard.
- The hazard has a significantly lower mean frequency of occurrence than other hazards with similar uncertainties, and will not result in consequences that are worse than those from other such hazards. The uncertainty in the frequency estimate for a hazard screened out in this manner is judged as not significantly influencing the total risk.

The quantitative criteria for screening out hazards should correlate with the core damage frequency from internal initiating events. Hazards of very low frequency but with potentially severe consequences in terms of releases of radioactive material should remain for the purposes of a Level 2 PSA.

The most important parameter relating to the damage potential of internal and external hazards is usually specified. However, several parameters should be specified if the damage potential of a hazard cannot be limited to consideration of a single parameter. All parameters specified for the hazards should be taken into account in performing the screening analysis (e.g. water level and pressure from the flow).

Initiating events occurring at the plant may be the result of the impact of a single hazard or of a combination of two or more hazards. When using the screening criteria, it should be justified that hazards whose combined impact can result in significant consequences are not excluded from further consideration, even though each of them, considered independently, would make a negligible contribution to risk. An example of such a combination of hazards is 'high winds' and 'external floods'. Even if each separate hazard could be screened out, the combination of hazards may have much higher impact on the risk to the plant, e.g. when 'external flood' is accompanied by or even caused by 'high winds'.

6.5 Bounding and detailed analyses

Hazards that survive screening should be considered in the framework of a bounding assessment and/or a detailed analysis.

This typically includes the following tasks:

- Collection of site and plant information supported, when feasible, by plant walk-downs.
- Hazard characterization: identification of hazards, calculation of hazard frequency and analyses of the impact of hazards.
- Integration of the Level 1 PSA for internal hazards with the Level 1 PSA for internal initiating events:
 - Determination of initiating events induced by the hazard;
 - Identification of necessary revisions of the Level 1 PSA for internal initiating events;
 - Analysis of specific dependences and common cause failures;
 - Analysis of specific data;
 - Analysis of specific human reliability aspects.
 - Qualitative and/or quantitative screening;
 - Quantification of the contribution of the hazards to core damage frequency (analysis of results, sensitivity studies, and uncertainty and importance analyses)
 - Documentation (with particular consideration given to the assumptions and references used in the analysis including quality assurance).

It is important to note that though initiating events caused by external and internal hazards are treated roughly with the same methodology as internal initiating events (IEs), they have important specificities that need to be emphasized, for instance:

- There are very important correlations between the IE, the component failures, and the containment behaviour;
- Particular components which are not considered in the case of internal IEs because they have a low failure probability, need to be introduced because their failure probability becomes significant in the case of internal or external hazards (electric cables in the case of fire, structures in the case of earthquake);
- Due to these specificities, the treatment of external and internal hazards needs particular analysis, including new event sequences and new systems analysis.

7 LOW POWER AND SHUTDOWN PSA

Learning objectives

After completing this chapter, the trainee will be able to:

1. Define plant operational states for which probabilistic safety assessment should be performed.
2. Describe the difference between PSA for full-power operation and PSA for low power and shutdown states.

7.1 General information

Recent studies have shown that the risk associated with low power and shutdown modes is comparable with and may be even higher than for full power conditions. This is explained by the fact that in contrast to full power operation, in low power and shutdown modes the operational configuration of the plant and conditions at the plant change significantly resulting in reduced numbers of physical and automated barriers:

- Reduced redundancy and diversity of mitigation systems;
- Increased reliance on operator actions;
- Increased susceptibility to fires and floods;
- Reduced containment / confinement integrity.

It is important that the whole operating cycle is covered in the PSA (see Figure 7.1); therefore low power and shutdown PSA is the essential element that should be included in the scope of the PSA overall in order to avoid missing contributors to risk from certain plant operational modes.

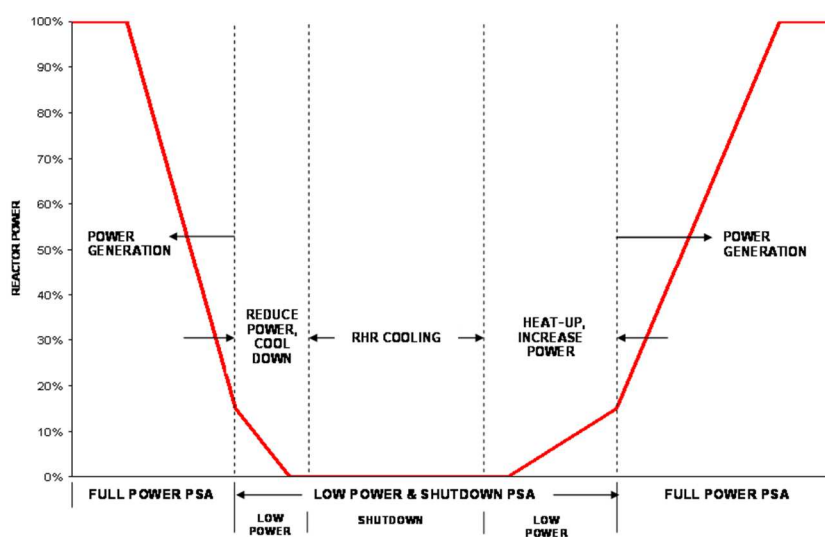


Figure 7.1: PSA coverage of plant operating modes.

In principle, Level 1 PSA for low power and shutdown states for internal initiating events is based on the same methodology as outlined for Level 1 PSA for full power states. However, the objective of the low power and shutdown PSA is not necessarily the determination of core damage frequency, since fuel damage frequency and inadvertent criticality may also be risk parameters of interest. There are also several important differences from full-power PSA that require special consideration in low power and shutdown PSA as discussed below [5, 26 - 29].

7.2 Plant operational states (POS)

Generally (for plants where refuelling is carried out off-line), there are three different types of outage as follows:

- Regular refuelling outages, during which major maintenance activities are also carried out;
- Planned outages, during which only specific maintenance activities are carried out;
- Unplanned but foreseeable outages that follow a disturbance during full power operation.

It is considered to be good practice to analyse all the types of outage mentioned above. The major specificity of the low power and shutdown PSA is that a large number and variety of plant configurations exist that would, if handled individually, lead to an excessive number of scenarios to be analysed. For dealing with the variety of plant states during low power and shutdown, a limited number of plant operational states are usually specified for which the plant status and configuration are sufficiently stable and representative.

The following physical and technical aspects of the plant states are typically taken into account when defining plant operational states:

- Reactor criticality (and/or shutdown margin);
- The level of decay heat;
- Temperature and pressure in the reactor coolant system;
- Water level in the primary system;
- Open or closed reactor coolant system;
- Operability status of loops in the reactor coolant system;
- Location of the fuel;
- Availability of safety systems and support systems, including consideration of whether they are controlled automatically or by manual actions;
- System alignments;
- Status of the containment integrity.

7.3 Initiating events analysis

There are certain features of low power and shutdown conditions that require special attention in PSA:

New or higher possibilities of initiators

Several types of initiating events should be considered in low power and shutdown PSAs:

- LOCAs caused by maintenance errors;
- Drain-down events caused by human error;
- Fuel pool leakage / drain-down;
- Inadvertent actuation of high pressure safety injection during cold states;
- Decrease in boron concentration/uncontrolled dilution;
- Return-to-criticality events, refuelling errors or errors in fuel handling;
- Direct fuel damage (e.g. heavy load drops into the reactor vessel, fuel pool or systems required to maintain critical safety functions);
- Fuel handling accidents (e.g. dropping of reactor fuel assemblies).

Thus the examination of plant procedures for changing RCS configurations, testing and maintenance is essential in the initiating events analysis for low power and shutdown PSA.

High potential for human-induced initiating events

These human errors are typically not quantified explicitly in full power PSA (e.g. included in initiating event frequencies), but should be quantified explicitly for shutdown modes.

Typical human induced IEs in shutdown conditions usually involve errors during operations, testing and maintenance and could lead to:

- Loss of cooling;
- Loss of inventory;
- Boron dilution;
- Fuel handling accidents.

Initiating events grouping in POSs

The groups of initiating events could be different in different POSs:

- Some initiating events may occur during many (or all) POSs;
- Some initiating events depend on specific POS (e.g. testing and maintenance, fuel handling activities).

Therefore each IE should be analysed for its applicability to a specific POS in the grouping process.

7.4 Reliability of systems, human reliability and data analysis

Important aspects of system modelling in low power and shutdown

PSA are as follows:

- Consistent accounting for POSs of specific system configurations such as initially available trains, operating / standby equipment, flow path alignments, success criteria, etc.
- Correct modelling of maintenance, such as correlations to POSs, correlations of equipment / trains / systems, consistent definitions of basic maintenance events, etc.

Errors that cause initiating events (type B) should be modelled explicitly in low power and shutdown PSA, in particular errors that involve with:

- System alignment changes;
- Transitions between plant operating states;
- Routine operations;
- Testing.

Post-initiator errors (type C) are generally quantified with the same methods as in full power PSA; however the following specific features for low power and shutdown PSA should be kept in mind:

- Conditions may develop slowly – difficult diagnosis;
- Long-time windows – some references recommend lower-boundary limits for cognitive errors if typical time-reliability correlations are used;
- Limited emergency procedures, training, and experience of shutdown events.

Table 7.1 below provides a comparative summary of aspects important for HRA in full power and shutdown conditions.

Table 7.1: Comparative summary of aspects important for HRA.

Full-Power PSA	Shutdown PSA
Short time windows ~0.5 hr	Long time windows ~0.5 - 10 hrs
Multiple alarms	Fewer alarms
Clearer indications	Less clear indications, temporary instrumentation
Symptom-based EOPs	Relaxed procedures
Simulator, classroom training	Limited training
Misdiagnosis generally considered difficult	More possibility of misdiagnosis
Most actions in MCR	Coordination of MCR and local actions
Human errors ~10% of CDF	Human errors ~80% of CDF

Low power and shutdown PSA requires collection of additional data to full power PSA:

- Frequency of each outage type (outage / year);
- Duration of each outage type (hours / outage);
- Duration of each plant operating state in each outage (hours /

- POS);
- Fraction of time in each POS;
- Shutdown maintenance data.

The last item is the most difficult data analyses task. It should be noted that generic data is not applicable to the shutdown maintenance data and plant-specific data for each outage should be collected.

8 PSA APPLICATIONS

Learning objectives

After completing this chapter, the trainee will be able to:

- 1. Describe the applications of probabilistic safety assessments for the assessment of overall plant safety.*
- 2. Describe the applications of probabilistic safety assessments in the context of a periodic safety review.*
- 3. Describe the applications of probabilistic safety assessments for security threats.*
- 4. Describe the applications of probabilistic safety assessments for the design evaluation.*
- 5. Describe the applications of probabilistic safety assessments in NPP operation.*
- 6. Describe the applications of probabilistic safety assessments for the evaluation of plant modifications.*
- 7. Describe the applications of probabilistic safety assessments for risk evaluations and for evaluation of regulatory decisions.*

There are several IAEA publications [5, 12, 13, 30] providing information on various PSA applications.

8.1 Safety assessment

Assessment of overall plant safety

The assessment of the overall plant safety represents the main purpose of performing PSA and includes identification and ranking of important design and operational features, of dominant accident sequences, systems, components, human interactions and dependences important for safety. A comparison of the results against safety goals or quantitative health objectives may be involved.

Periodic safety review

PSA provides useful insights to support a periodic safety review. A safety assessment process consists in identifying safety issues, determining their safety significance and making decisions about the need for corrective measures. A major benefit of including PSA in periodic safety reviews is the creation of an up-to-date overview of the whole plant. PSA may help in identification of real cost-effective improvements to safety. The important issues in this application are the use of plant-specific data, modelling of as-built-as-operated plant conditions, and addressing the possible impact of ageing phenomena and component lifetime considerations on the overall risk probability.

Analysis of the degree of defence against assumed terrorist attack scenarios (security application)

This assessment may include identification of vital plant areas which pose a significant risk to the plant in case of malevolent acts, and of

exploratory design options aimed at reducing the risk. After September 11, 2001, this application was introduced to support analysis of the protection of NPP against similar attack scenarios. Other types of terrorist attack scenarios, e.g. external explosions, can be considered within the scope of this application.

8.2 Design evaluation

Application of PSA to support decisions made during the NPP design

This application focuses on the evaluation of whether the design of safety features is balanced and there is no disproportionately high contribution from a particular initiating event or an accident sequence. Identification of design weaknesses and effective areas for improvement in view of plant risk is also an important area of application. The assessment may include investigation of variants and exploratory design options, sufficiency in systems' redundancy and diversity, effectiveness in emergency and accident management measures, as well as development of reliability and availability targets for SSCs to meet safety goals, if set.

For example, the PSA can be used as a supporting tool to select or modify the design basis accidents and design extension conditions [3], to determine the classification of safety-related SSCs, to define general design criteria, and to develop SSC reliability and availability targets.

Assessment of the safety importance of deviations between an existing plant design and updated/revised deterministic design rules

The assessment may include investigation of the risk-significance of deviations from revised design rules, often performed in the framework of a periodic safety review. The key issue in this application is the possible impact of changes in the design rules on risk associated with plant operation. For example, for operating plants constructed in accordance with the old design rules, PSA results and risk probabilities can be used to justify the low risk significance of certain deviations from revised rules.

8.3 NPP operation

NPP maintenance

Maintenance programme optimization: The application includes assessment, optimization and establishment of maintenance plans and procedures in relation to plant risk. Maintenance activities are assessed to assure that risk significant systems and equipment are being adequately maintained to support the required reliability, and that maintenance activities do not reduce plant safety and increase risk

by, for example, increased equipment unavailability resulting from extensive maintenance. The focus is on equipment with the greatest impacts on plant risk. Opportunities for reduced or eliminated maintenance tasks are defined and evaluated for SSCs that have low risk significance and for maintenance that does not affect critical functions of the SSCs.

Risk-informed house-keeping: This application is intended to assure low risk contributions from external events (e.g., seismic) and internal hazards (e.g., fire, floods) by directing housekeeping activities to areas important to risk.

Risk-informed support for the plant ageing management programme: The aim of this application is to optimise the scope of the plant-specific ageing programme for safety-related equipment. It includes identification of safety significant components and may involve modelling of ageing effects in PSA and identification of the risk significant SSCs potentially degrading due to ageing phenomena.

Accident mitigation and emergency planning

Development and improvement of emergency operating procedures: The systematic assessment of plant vulnerabilities and the insights derived from the PSA process are used to establish or improve the EOPs by providing assurance that a broad range of vulnerabilities is addressed in a realistic, appropriately detailed and consistent manner. The integral view of accident progression provides information on the benefits and drawbacks of various operations in abnormal plant states. Typically, accident sequence analysis in PSA is carried out using existing EOPs and assessment of associated human interactions. This in turn provides detailed information for reconsidering EOPs and eventual improvements in the light of PSA insights. PSA can also provide the basis for specifying the decision points at which transition into the SAMG phase should occur.

Support for severe accident prevention: Existing, alternative or additional systems, equipment and measures are evaluated and implemented in the accident management procedures with the purpose of restoring the function of safety-related systems and for preventing degradation of events into severe accidents.

Support for severe accident mitigation: PSA helps to understand accident progression, identification of success paths and associated strategies, prioritising safety features to reduce risks. The integral view of plant response utilized in PSA methodology is helpful in discerning the potential for negative effects of certain measures.

Support for NPP emergency planning: Based on PSA, important elements of emergency planning are explored and appropriate strategies are developed. The issues to be explored are the characteristics of the plant, the plant site and the different possible

countermeasures (such as sheltering, evacuation, iodine prophylaxis, long-term relocation, land decontamination, food bans). This requires a Level 3 PSA. Specific applications include possible adjustments to the emergency planning zones (EPZ), refinement of emergency action levels, and focusing the resources for evacuation and sheltering.

Personnel training

Improvement of operator training programme: PSA is used to improve the operator training programme by providing information on accident processes, the relative likelihood of the dominant accident sequences, and the associated operator actions required to prevent or mitigate core damage.

Improvement of maintenance personnel training programme: Training of maintenance staff is enhanced based on insights and information from the PSA. This results in an increased focus on risk-significant SSCs and on risk-significant functions and failure modes that must be addressed in the maintenance programme, as well as opportunities to optimize maintenance tasks that are not significant to risk management.

Improvement of plant management training programme: This includes adequate communication of the techniques, applications and implications of PSA to plant management in order to develop an integral understanding in terms of management responsibilities.

Risk-based configuration control/ Risk Monitors: This application deals with maintenance planning and assesses the risk associated with taking out of service several safety important components simultaneously. This is a forward-looking application of PSA consisting of supporting the preparation, planning and scheduling of plant activities and component configurations. This application can be performed with an on-line or off-line PSA model. An example is the use of the Risk Monitor tool to evaluate the time-dependent risk profile for a future time period, in which a series of plant configuration changes is being planned.

Real time configuration assessment and control (response to emerging conditions): This application deals with risk assessment and the follow-up involves the online use of the Risk Monitor model by plant personnel in order to keep the risk due to actual configurations, plant activities and unanticipated events at an acceptable level.

Exemptions to Technical Specification (TS) and justification for continued operation: As a comprehensive tool describing the risk associated with a particular plant configuration, the PSA can provide useful support to TS exemption justifications and/or to proposals for mitigatory or compensatory measures, or to justify the relevance of these measures. This application deals with temporary changes and hence the decision criteria may be less restrictive than for the case of

permanent changes because of the one-time nature of the exemption.

Dynamic risk-informed TS: Dynamic risk-informed TS relaxes some of the rigidly based TSs on the integral view of a PSA. Allowed Outage Times (AOTs) are calculated for each plant state taking into account the complete picture of plant configuration and equipment out-of-service combinations. In some cases the calculated AOT may be shorter than the standard fixed technical specification version and in some cases a longer AOT is justified, but in all cases a careful evaluation is made at all times of the complete picture of plant configurations, including safety and non-safety related SSCs.

8.4 Permanent changes to the operating plant

Plant changes

NPP upgrades, back-fitting activities and plant modifications: Identification of weaknesses and effective areas for improvement in plant design and operational features in view of plant risk. PSA arguments are used to support the selection, design, implementation, justification, and licensing of plant upgrades.

Life time extension: This application is often referred as a sub-case of the periodic safety review with consideration of ageing effects beyond the design lifetime. It involves modelling ageing effects in the PSA.

Technical specification changes

Determination and evaluation of changes to allowed outage time and changes to required TS actions: Required actions in TS have been typically derived on a classical engineering basis. Some items can be re-evaluated based on the PSA and eventually changed according to their risk-significance. In addition, the PSA may be used to support the optimization of maintenance tasks with respect to whether they must be done during outages or whether on-line maintenance is appropriate.

Risk-informed optimisation of TS: The TSs define limits and conditions for operation, testing, and maintenance activities as a way of assuring that the plant is operated safely. From time to time, the plant operator may need a TS exemption due to operational burdens and constraints. This application is used to optimise TS provisions.

Determination and evaluation of changes to surveillance test intervals: PSA based evaluation of surveillance test intervals (STIs) considers the risk from unavailability due to undetected failures, and the risk from unavailability due to tests and test induced failures. The goal is to optimize the STIs with respect to their impact on equipment reliability and how these tests impact the cost of operations.

Risk-informed in-service testing: Use of PSA to support the IST

programme, taking into account the relative risk significance of the components to be tested, is the focus of the application. The relative risk significance is assessed using a blend of probabilistic and deterministic methods before any test interval is changed and the aggregate impact of the changes is evaluated. This results in relaxation of testing requirements for low risk significant SSCs and SSC functions, and increased requirements for higher risk significant SSCs, yielding an optimized testing programme.

Risk-informed in-service inspections: The risk-informed in-service inspection (RI-ISI) methodology consists of ranking the elements for inspection, such as welds in piping systems, according to their risk significance and developing an inspection strategy (frequency, method, sample size, etc.) commensurate with their risk significance. It provides a framework for effective allocation of inspection resources and helps to focus the inspection activities where they are most needed. In addition, an understanding of the most likely degradation mechanisms is developed, which is used to focus the required inspections on using the most appropriate inspection methods for the anticipated damage mechanisms.

Establishment of graded QA programme for SSC

Equipment risk significance evaluation: The PSA provides the necessary insights that can be used to determine the relative safety significance of plant equipment. These probabilistic insights are, for example, utilized to help identify low/high safety significant SSCs that are candidates for reductions/improvements in QA treatment.

Evaluation of the risk impact of changes to QA requirements: Changes in QA treatment of SSCs are investigated or explored within the framework of the PSA. Sensitivity studies are required to assure robust decision making. Change in risk probabilities are used to determine the risk significance and risk acceptability of the proposed change. An example is evaluating the change in risk associated with relaxation of QA requirements.

8.5 Supervisory activities

Performance monitoring

Planning and prioritization of inspection activities (regulatory and industry): PSA-based ranking of design and operational features is used to focus resources for regulatory and industry inspections on important issues and equipment.

Long-term risk-based performance indicators: The long-term risk based indicators focus on monitoring plant behaviour in order to get insights into the past history of NPP safety and to update the calculated average CDF. Long-term risk-based indicators can help to pinpoint ageing effects on components and systems. For long-term

planning, the assumptions regarding planned design changes, expected component behaviour, etc. can be introduced in the PSA models and data, and can be analysed to obtain the expected average CDF for the next period.

Short-term risk-based performance indicators: Risk-based indicators for short-term use require instantaneous evaluation of risk. This type of application provides information on changes in CDF due to plant events and risk associated with planned activities.

Performance assessment

Assessment of inspection findings: This type of application provides information on changes in risk measures associated with inspection findings. Change in risk and conditional risk probabilities can be used to evaluate the risk impact of degradations or safety issues that are found during inspections and to evaluate possible corrective actions.

Evaluation and rating of operational events: From PSA-based extrapolation of operational events to accident scenarios with serious consequences, valuable insights can be gained regarding accidents on the basis of minor incidents, without suffering their real consequences. PSA can be used to analyse plant events which may initiate a plant trip, degrade or disable safety systems, or both simultaneously.

8.6 Evaluation of safety issues

Risk evaluation

Risk evaluation of corrective measures: Based on PSA insights corrective measures regarding safety issues are developed. This may include exploratory investigation on different variants to resolve a particular issue.

Risk evaluation to identify and rank safety issues: As a result of a PSA, important new plant-specific safety issues and generic issues may be identified. Furthermore, PSA is used for evaluating the relative importance of existing and new safety issues. Also safety issues identified outside the PSA can be evaluated by the PSA to determine their risk significance once the issues have been assessed for risk characterization.

Regulatory decisions

Long-term regulatory decisions: PSA insights are used to guide long-term prioritization of regulatory objectives and requirements, and of related safety research. PSA results are used to develop risk insights, and strategies to maintain or reduce risk levels are devised. Change in risk probabilities are used to evaluate possible changes to requirements needed to implement the risk management strategy.

Interim regulatory decisions: PSA is used to alleviate a regulatory

concern while longer-term solutions can be evaluated. Issues that typically require an interim decision are: (a) the need for regulatory action in response to an event at a plant, (b) one-time exemptions from TS or other licensing requirements, and (c) temporary modifications to hardware configuration or procedures.

9 QUESTIONS

1. What is the main purpose of performing safety analyses?
2. Whose responsibility is the performance of safety analyses?
3. What are the main goals/outcomes of probabilistic safety assessments?
4. What is the scope of probabilistic safety assessments and what are the levels of probabilistic safety assessments?
5. Name the plant states for which probabilistic safety assessments are performed.
6. Which initiating events and hazards have to be taken into account when performing probabilistic safety assessments?
7. What are PSA Level 1 objectives?
8. Explain the concept of fault trees and event trees.
9. Explain the concept of common cause failures and human reliability analysis.
10. What are PSA Level 2 objectives?
11. What are the general steps undertaken when performing Level 2 PSA?
12. Describe briefly the PSA Level 1 to PSA Level 2 interface.
13. Describe accident progression and containment analyses at PSA Level 2.
14. Describe in simple words the source term.
15. What are PSA Level 3 objectives and what are the main steps needed for performing PSA Level 3?
16. Why do we need atmospheric dispersion and deposition models?
17. How do we use the meteorological data, exposure pathways and data on population, agriculture and economy for dose assessment?
18. What is the difference between internal and external hazards?
19. Name a few applications of probabilistic safety assessments.

10 REFERENCES

- [1] INTERNATIONAL ATOMIC ENERGY AGENCY, IAEA Safety Glossary; Terminology Used in Nuclear Safety and Radiological Protection, 2007 Edition, IAEA Vienna (2007).
- [2] EUROPEAN ATOMIC ENERGY COMMUNITY, FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS, INTERNATIONAL ATOMIC ENERGY AGENCY, INTERNATIONAL LABOUR ORGANIZATION, INTERNATIONAL MARITIME ORGANIZATION, OECD NUCLEAR ENERGY AGENCY, PAN AMERICAN HEALTH ORGANIZATION, UNITED NATIONS ENVIRONMENT PROGRAMME, WORLD HEALTH ORGANIZATION, Fundamental Safety Principles, IAEA Safety Standards Series No. SF-1, IAEA, Vienna (2006).
- [3] INTERNATIONAL ATOMIC ENERGY AGENCY, Safety of Nuclear Power Plant: Design, Safety Standards Series No. SSR-2/1, IAEA, Vienna (2012).
- [4] NUCLEAR REGULATORY COMMISSION, Reactor Safety Study: An Assessment of Accident Risks in US Commercial Nuclear Plants - WASH-1400-MR, NUREG 75-014, Washington, DC (1975).
- [5] INTERNATIONAL ATOMIC ENERGY AGENCY, Development and Applications of Level-1 PSA, Safety Standards Series No. SSG-3, IAEA, Vienna (2010).
- [6] INTERNATIONAL ATOMIC ENERGY AGENCY, Procedures for Conducting Probabilistic Safety Assessment of NPP Level 1. Safety series No. 50-P-4, IAEA, Vienna (1992).
- [7] NUCLEAR REGULATORY COMMISSION, Fault Tree Handbook, NUREG-0492, Washington, DC (1981).
- [8] WIERMAN, T., et al., Common-Cause Failure Event Insights (NUREG/CR-6819), Washington, DC (2003).
- [9] OECD NUCLEAR ENERGY AGENCY, ICDE Project Report: Collection and Analysis of Common-Cause Failures of Switching Devices and Circuit Breakers, NEA/CSNI/R(2008)1, OECD/NEA, Paris (2008).
- [10] SWAIN, A.D., GUTTMAN, H.E., Handbook for Reliability Analysis with Emphasis on Nuclear Power Plants Applications, NUREG/CR-1278, Washington, DC (1983).
- [11] FORESTER, J., et al., Evaluation of Human Reliability Analysis Methods Against Good Practices, NUREG-1842, Washington, DC (2006).
- [12] INTERNATIONAL ATOMIC ENERGY AGENCY, Determining PSA Quality for various applications, TECDOC 1511, IAEA, Vienna (2007).
- [13] INTERNATIONAL ATOMIC ENERGY AGENCY, Development and Applications of Level-2 PSA, Safety Standards Series No. SSG-4, IAEA, Vienna (2010).
- [14] KHATIB-RAHBAR, M., CAZZOLI, E., LEE, M.,

- NOURBAKSH, H., DAVIS, R., SCHMIDT, E., A probabilistic approach to quantifying uncertainties in the progression of severe accidents, Nucl. Sci. Eng. 102 (1989) 219.
- [15] BUDNITZ, R.J., APOSTOLAKIS, G., BOORE, D.M, CLUFF, L.S, COPPERSMITH, K.J., CORNELL, C.A., MORRIS, P.A., Recommendations for Probabilistic Seismic Hazard Analysis: Guidance on Uncertainty and Use of Experts, Rep. NUREG/CR-6372, Lawrence Livermore National Laboratory, Livermore (1997).
- [16] MEYER, M.A., BOOKER, J.M, Eliciting and analyzing expert judgment: A practical guide, Rep. NUREG/CR-5424, Los Alamos National Laboratory, Los Alamos (1990).
- [17] THEOFANOUS, T., YAN, H., “ROAAM: A risk-oriented accident analysis methodology”, Proc. Intl. Conf. on Probabilistic Safety Assessment and Management, Beverly Hills, 1991, Elsevier Science Ltd, New York (1991) 1179.
- [18] HARPER, F.T., et al., Evaluation of Severe Accident Risks: Quantification of Major Input Parameters, Rep. NUREG/CR-4551, Vol. 2, Part 4, Sandia National Laboratories, Albuquerque (1991).
- [19] MENDOZA, Z.T., FREEMAN, M., LEONARD, M., EUTO, J., HALL, J., Generic Framework for IPE Back-End (Level 2) Analysis, NSAC-159, Electric Power Research Institute, Palo Alto, CA (1991).
- [20] REMPE, J.L., et al., Light Water Reactor Lower Head Failure Analysis, Rep. NUREG/CR-5642, Idaho National Engineering Laboratory, Idaho Falls (1993).
- [21] CHU, T.Y., PILCH, M.M., BENTZ, J.H., JUDWIGSEN, J.S., LU, W-Y, HUMPHRIES, L.L., Lower Head Failure Experiments and Analyses, Rep. NUREG/CR-5582, Sandia National Laboratories, Albuquerque, (1998).
- [22] BREITUNG, W., CHAN, C.K., DOROFEEV, S.B., EDER, A., GELFAND, B.E., HEITSCH, M., KLEIN, R., MALLIAKOS, A., SHEPHERD, J.E., STUDER, E., THIBAULT, P. Flame Acceleration and Deflagration-to-Detonation Transition in Nuclear Safety, State-of-the-Art Report by a Group of Experts, NEA/CSNI/R(2000)7, OECD Nuclear Energy Agency, Paris (2000).
- [23] GIESEKE, J.A., CYBULSKIS, P., JORDAN, H., LEE, K.W., The Source Term Code Package, Proc. International symposium on source term evaluation for accident conditions; Columbus, OH (USA); 28 Oct - 1 Nov 1985, p. 675-685, ISBN 92-0-020086-9, International Atomic Energy Agency, Vienna (1986).
- [24] OECD NUCLEAR ENERGY AGENCY, Evaluation of Uncertainties in Relation to Severe Accidents and Level-2 Probabilistic Safety Analysis, Workshop Proceedings, Aix-en-Provence, 7-9 November 2005, NEA/CSNI/R(2007)2, OECD/NEA, Paris (2007).
- [25] INTERNATIONAL ATOMIC ENERGY AGENCY, Procedures for Conducting Probabilistic Safety Assessment of Nuclear

- Power Plant Level 3, Safety Series No. 50-P-12, IAEA, Vienna (1996).
- [26] INTERNATIONAL ATOMIC ENERGY AGENCY, Probabilistic Safety Assessments of Nuclear Power Plants for Low Power and Shutdown Modes, IAEA-TECDOC-1144, IAEA, Vienna (2000).
 - [27] AMERICAN NUCLEAR SOCIETY (ANS), Low Power and Shutdown PRA Methodology, ANS Standard 58.22 (Draft; in development).
 - [28] NUCLEAR REGULATORY COMMISSION, Evaluation of Potential Severe Accidents during Low Power and Shutdown Operations at Surry, Unit 1", NUREG/CR-6144, Washington, DC (1995).
 - [29] NUCLEAR REGULATORY COMMISSION, Evaluation of Potential Severe Accidents during Low Power and Shutdown Operations at Grand Gulf, Unit 1", NUREG/CR-6143, Washington, DC (1995).
 - [30] INTERNATIONAL ATOMIC ENERGY AGENCY, Applications of Probabilistic Safety Assessment (PSA) for Nuclear Power Plants, IAEA-TECDOC-1200, IAEA, Vienna (2001).

The views expressed in this document do not necessarily reflect the views of the European Commission.